

THE ENERGY BALANCE PROJECT

Implementation, Preservation and Evidence

Part III

Revised Gaussian Explanatory Edition

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What exists, and what is proposed

This volume explains how a mathematical account becomes a testable scientific system. It also explains how to avoid reporting a proposed system as though it already existed. Those are inseparable tasks: the credibility of an implementation depends on knowing exactly what it implements and what its tests actually establish.

The project has historical implementations and experiments under earlier potential and controller definitions. It also has a prospective Local Gaussian programme, preserved candidate code, and source-locked framework components on other Git lineages. These are not one interchangeable engine. The active chapters identify their differences. A historical appendix retains the original D1–D10, quote-validation, adversarial and service-alignment chapters without rewriting their results under the Gaussian formula.

The central reading rule

In this book, a prospective interface is not an implemented service; a passing arithmetic fixture is not a scientific campaign; and a historical result is evidence only for its own registered model and conditions.

This edition follows Part II's revised numbering, beginning at Chapter 47. Original chapter numbers in the historical appendix remain archival labels. The supplied original Part III PDF is preserved. New chapters are editable reconstructed source, retaining the original teaching style rather than claiming access to the original typesetting manuscript.

No Gaussian runtime, scientific trajectory or new experiment was executed to write this book. The prospective chapters describe responsibilities and acceptance boundaries, not accepted runnable APIs or a sealed study packet. Parameters, thresholds, seed families, horizons and cloud configurations are not selected here. Historical results are not erased because their model is no longer the active candidate.

Reading the software and the evidence

Chapters 47–55 follow one proposed scientific event from its declared state through forcing, candidates, physical screening, valuation, affordability, selection, execution records and reproducibility. The ordering is the new programme’s intended profile; reconciliation with the existing framework remains an explicit dependency, not a completed migration.

Chapters 56–57 explain the historical architecture and evidence. They are the bridge to the unaltered historical pages at the back. Chapter 58 examines measurement and the open loss-accounting boundary. Chapter 59 asks what a long-run result could mean. Chapter 60 supplies a worked audit and a precise map of the next separately authorized stages.

A reader interested in implementation should ask three questions of every module: which facts may it read, which facts may it write, and what exact claim does its output support? A reader interested in evidence should add: which version ran, what comparison was fixed beforehand, and where are the complete records, including refusals and negative outcomes?

The historical appendix reproduces publication-era prose. Words such as “current” there refer to that source’s time, not to this revision. The later finalized Gate 1D-C record is treated separately in Chapter 57. Neither the appendix nor that later record demonstrates success of the new Gaussian random-affordable programme.

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From a law to an executable claim

A mathematical equation can fit on one line while its faithful implementation requires many boundaries. The equation does not specify how state is frozen, how an action is identified, which module may read another cell, or what happens if a record is written twice. These are not decorative engineering details. They determine whether the program computes the object named in the theorem.

47.1 Four kinds of completion

A derived identity is mathematically complete when its statement and proof have the required assumptions. A component is implemented when its actual code performs its declared function. A test is completed when it has executed and produced inspectable output. A scientific study is completed only under its registered execution and finalization contract. A file named `runner.py` establishes none of the last three by its name alone.

The current Gaussian direction is prospective [3]. The repository contains historical executable engines, preserved candidate harness code and framework objects on source-locked lineages. Their presence does not establish an accepted Gaussian runtime. This chapter uses “would” and “must” for prospective responsibilities, not as euphemisms for completed implementation.

47.2 A module’s scientific contract

For each component, state its inputs, permitted reads, outputs, permitted writes, refusal conditions and version identity. A value evaluator should not choose the action. A physical checker should not improve an action’s score. A selector should not alter quantities after valuation. A logger should not repair a failed invariant by changing history.

For example, an evaluator might receive a frozen state projection, reference and scale declarations, and a finite accepted group. It would return the exact field value and declared attribution record without changing physical state. An execution adapter would later apply the exact selected group, under separate permission. Keeping those calls separate makes it possible to test arithmetic without accidentally running a model.

47.3 Why a generic framework is still useful

The source-inspected F2 decision [6] identifies reusable identity, canonical serialization, typed state/action/support records, information contracts, provenance and ledger structures. It does not say the whole framework already fits the Gaussian programme unchanged.

In particular, its potential declaration is metadata, not a numerical Gaussian evaluator. Its named scientific advancement routes are deliberately guarded. Its physical service-capacity records are not earned EBU balances. Reusing those names while changing their meanings would conceal a new scientific model inside an apparently routine software integration.

47.4 No framework replacement by default

The smallest honest path is to reuse inspected generic pieces and add separately versioned adapters where the active scientific meaning differs. Wholesale copying can import historical assumptions; wholesale replacement can discard proven provenance discipline. The planning decision is neither “everything works already” nor “nothing can be reused.”

The source locks matter. Some relevant framework files live on another lineage and are absent from the active checkout. An ignored bytecode directory is not an authoritative source package. A later implementation task must inspect the exact committed files it intends to reuse, rather than infer availability from import names or an old chat summary.

Worked review question

A function named `advance_epoch` exists but always refuses because its scientific execution permit is unavailable. Is epoch advancement implemented? The refusal guard is implemented. Successful scientific advancement is not established. Both facts belong in the status report; neither “there is no software” nor “the engine is ready” is accurate.

Chapter recap

The bridge from mathematics to software is a set of typed responsibilities and verifiable effects. The bridge from software to evidence requires an additional registered study, not a stronger adjective in the implementation description.

The smallest honest Gaussian world

The first active world should be small enough that every represented quantity has an unambiguous meaning. Small does not mean vague. A two-cell model with mixed units and hidden replenishment is less interpretable than a larger model with explicit boundaries.

48.1 Physical state and reference are different records

The stock vector x changes when physical actions or forcing occur. The reference x^* and scales σ are fixed configuration in the first domain. Both state and reference are nonnegative and share total mass M ; every scale is finite and strictly positive. NaN, infinity, a zero scale or a negative stock is not a small deviation to smooth away.

The potential is

$$V(x) = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2. \quad (48.1)$$

It is a derived value, not another physical carrier. Storing a cached value may be convenient, but replay must be able to recompute it from the exact declared state and configuration. Updating the cache without the state, or the state without invalidating the cache, risks valuing a nonexistent world.

48.2 Accounts belong beside, not inside, the stock vector

The proposed augmented record includes physical state x , persistent capacity B , signed external audit J , and random-stream replay coordinates. They have linked versions but distinct meanings. Adding an EBU balance to a water total is dimensionally invalid. Treating a physical capacity constraint as a spendable balance is semantically invalid even if both are numbers.

The old candidate C3 export budget also used a letter resembling B . That does not make it the new persistent account. A safe specification uses explicit type names and conversions where necessary, not a shared generic field called budget for every purpose.

48.3 Atomic action declarations

An action records its identity, source, destination, finite quantity, carrier, write support and relevant constraints. For a lossless transfer, $\delta_a = q_a(-e_i + e_j)$. A group retains every child and its accepted increment; it also records the aggregate δ_G .

The source-side quantity convention must remain explicit when importing historical rate-based examples. A duration belongs in the conversion $q = \Delta t j$, not in both the conversion and the finite evaluator. Non-unit-duration fixtures expose errors that unit-duration examples hide.

48.4 What is deliberately absent

The active first domain contains no Allee regeneration, lower/upper homeostatic band, hidden demand-serving controller or autonomous gradient flow. Its reference is not an empirically fitted stationary mean by default. Its scales are not automatically measurement-error bars. It contains one homogeneous conserved scalar, not an uncalibrated mixture of energy, water, medicine and welfare.

This is a scope statement, not a claim those phenomena are unimportant. Adding them later changes state, dynamics or valuation and must receive its own derivation and registration. An honest small world provides a controlled question; an underspecified broad world provides ambiguous answers.

A configuration audit

Suppose a proposed file gives $x = (18, 2)$, reference $(10, 10)$ and scales $(2, 2)$. It passes the elementary domain checks. Replace the reference with $(12, 12)$ and closed mass compatibility fails. Replace a scale with zero and valuation is undefined. Add a regenerative source label without a regeneration law and the label has supplied no physical mechanism.

Each failure should identify the exact predicate and field. “Invalid configuration” alone is difficult to audit. Silently normalizing the reference or clipping a stock would make the accepted model differ from the author’s declaration.

Chapter recap

Typed state, fixed geometry and complete action declarations make later checks meaningful. The first world is intentionally limited; its limitations must remain readable rather than filled with inherited historical defaults.

An event begins with its baseline

The same physical increment can receive different values at different states. It follows that the event boundary is part of the scientific definition, not merely a scheduler detail. A record must identify when forcing occurs, when the state is frozen and which version each candidate uses.

49.1 The intended forcing-first order

The new programme proposes the following logical sequence:

Phase	Responsibility
1	Apply the declared external forcing and record its signed audit.
2	Freeze one post-forcing physical baseline and its configuration.
3	Generate EBU-blind candidates/groups and check joint physical feasibility.
4	Evaluate exact group value and declared path attribution.
5	Aggregate by declared owner and screen affordability.
6	Make an unweighted random choice from the declared eligible menu.
7	Execute exactly that selected accepted vector, then record accounting effects once.

These are explanatory phases, not a new accepted enumeration of framework phase constants. The inspected framework has its own fixed ten-phase chronology, including natural drive in a different position. A recognized Gaussian event profile must resolve that difference before runtime adoption [6].

49.2 Why forcing is separate

Write x_t for the pre-forcing state and z_t for the post-forcing baseline. The external record is $D_{\text{ext},t} = V(z_t) - V(x_t)$. All candidates in that epoch compare

z_t with their own declared endpoints. An idle actor therefore has zero action value even if forcing just improved the field.

If one candidate uses x_t and another uses z_t , their comparison includes different external effects. If forcing changes again between quote and execution, the committed conditions need explicit treatment; the software cannot quietly call the new state the old baseline.

49.3 Epoch identity binds meaning

A useful epoch commitment identifies the state version, physical profile, potential and parameter version, accepted child vector, complete support, path semantics, owner-rule version and numerical policy. Not every module needs to read all of these values, but the composite record must connect them unambiguously.

Changing an accepted quantity changes the endpoint and generally every shared-path attribution. Removing one child can change the value of the others. Such changes require a new affected valuation and authorization record, not reuse of an old number whose identity happens to look familiar.

49.4 A worked baseline error

At reference (10, 10) the potential is zero. Forcing produces (14, 6) with potential four. A restoring action ends at (12, 8) with potential one. The actor's value is $4 - 1 = 3$. Comparing the original reference directly with the endpoint gives $0 - 1 = -1$, which includes the shock and answers a different question. Both subtractions are arithmetically correct; only one is the declared actor account.

This is why a test of subtraction alone is insufficient. The implementation must also test which state commitment supplies each operand.

49.5 Failure and publication order

A prospective runtime must specify what is durable before state advancement becomes possible, what constitutes one completed event and how partial publication is detected. Book prose cannot substitute for that execution contract. Historical Gate 1D-C has a particular durability protocol; it must not be copied as though every new study shares its exact permissions.

Chapter recap

The baseline, chronology and event identity are part of the model. A new diagram explains the intended process but does not amend the framework's existing scientific order by itself.

Candidates, physical permission and group selection

The candidate menu is where an apparently innocent implementation detail can become an undeclared controller. If the generator proposes only actions that point toward reference, the later selector is not testing unranked physical opportunities, however random its final choice may be.

50.1 EBU-blind does not mean physically ignorant

The proposed generator may use the physical information permitted by its contract: topology, available stock and declared capacity constraints, for example. It must not rank or shape opportunities using EBU, global potential, future recovery or an undeclared service objective. Exactly which local physical information it may read remains a declared design choice.

A finite menu restricts the experiment. If physical quantities are continuous but the menu contains only a few values, the study concerns that menu, not all possible divisible action. This is acceptable when explicit. It becomes misleading when a failure to reach reference is generalized without checking whether the menu ever made the necessary quantity available.

50.2 Joint constraints come before value

The physical layer validates a complete group. It checks nonnegative end-points, conservation, aggregate source and destination constraints, allowed edges and declared timing. It uses the same rules for the EBU arm and the physical-random control.

Two individually valid children may exceed a shared cap. If a resolver changes their quantities, the new accepted vector is the one to value. After selection, execution must not run another hidden resolver and keep the old quote. That would settle one vector while performing another.

The order “physical permission before value” does not mean physical permission chooses the most valuable allocation. A proportional allocator, priority allocator and refusal rule are different policies. A source-locked study must say which one it uses or avoid unresolved contention in its declared domain.

50.3 Uniform over what?

Suppose a menu contains physical options A and B, but A appears under two identifiers. Uniform sampling over entries gives A probability two-thirds; uniform sampling over unique physical options gives one-half. Neither interpretation can be inferred from the word random alone.

Group size matters too. Uniform sampling of groups can give an action that belongs to many groups a high chance of execution. Uniform sampling of owners first produces another distribution. A no-action option can affect refusal and time occupation even when it has zero field value. All of these choices change the transition kernel and must be fixed prospectively.

50.4 Local calculation and global assembly

An exact group value can be evaluated from its affected factors, while the group assembler still reads a world-wide menu. Thus factor-local valuation does not prove a decentralized actor process. The specification must name the chooser, its view, its communication and its authority over owners.

There is no need to settle every distributed-system research question in the first model. It is necessary to state what was actually chosen. A centralized experimental sampler can be studied honestly if it is not advertised as a proof of fully local institutional implementation.

A useful negative control

Consider two baseline states with identical permitted local observations but different hidden remote potential. The candidate generator should not change its output merely because the hidden diagnostic changed. Designing such an information-boundary fixture is useful; executing a full transition with it belongs to the separately authorized integration stage.

Chapter recap

The menu, duplicate policy, group assembly and sampling unit are part of the science. They should not be left to incidental list construction or silently optimized in the name of convenience.

An exact evaluator and a trustworthy quote

An evaluator’s task is narrow: given a declared baseline and accepted finite action, return the corresponding value without moving the model. A narrow task is easier to audit because unexpected side effects have no legitimate place to hide.

51.1 Independent arithmetic routes

For the Gaussian domain, two direct routes should agree:

$$E_G = V(z) - V(z + \delta_G), \quad (51.1)$$

$$E_G = -\mu(z)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G. \quad (51.2)$$

Local affected-factor evaluation provides a third comparison with the full diagnostic difference. These routes share a mathematical specification but can expose different implementation mistakes: wrong signs, omitted curvature, incomplete support or a stale baseline.

A test that compares a function with another call to the same function is not an independent check. A fixed rational example with an independently written expected value is often more revealing than thousands of repeated calls through one mistaken helper.

51.2 Exact algebra and numerical representation

The formula is exact mathematics. Floating-point evaluation is not exact real arithmetic. A future protocol must specify whether representation is rational, fixed-point or floating-point, and what refusal or tolerance rules apply near zero, affordability and conservation boundaries.

Reporting a residual of zero on selected fixtures does not prove all real inputs will produce exact machine zero. Conversely, choosing a convenient tolerance after observing a failure can weaken the scientific contract. The numerical policy must be fixed before outcomes can influence it.

51.3 Value, attribution and settlement have different outputs

The group value is a single field difference. A common-path evaluator can add R_a for each child and verify $\sum_a R_a = E_G$. A settlement service would additionally require an accepted owner/policy map and a valid event identity. Combining all three in one return value called reward invites the caller to assume an authority that the evaluator never established.

The preserved candidate code already distinguishes physical constant-rate path semantics from endpoint interpolation. That distinction is reusable as an idea. Its mere presence in a candidate module does not authorize spendable Gaussian capacity or prove every interpretation guard sufficient.

51.4 A quote must describe what was accepted

For a one-dimensional accepted schedule, the record can identify the full function on its accepted domain. A measured shortfall can select a point on that precommitted function. For a simultaneous group, changing one child may alter every path attribution. The applicable multidimensional or re-epoching contract must therefore be declared rather than extrapolated from the simplest single-action schedule.

An overshoot outside a committed domain is a protocol violation. It does not justify an improvised positive credit, nor an arbitrary punishment presented as physical EBU. Physical measurement, account calculation and sanctions remain separate questions.

Worked evaluator audit

At (18, 2), reference (10, 10), scales (2, 2) and quantity four, the endpoint is (14, 6). The potential route gives $16 - 4 = 12$. The tangent is sixteen and the curvature correction four, again giving twelve. A returned value of sixteen suggests the finite evaluator has silently become a first-order approximation. A returned value of fourteen for two two-unit children suggests independent singleton values were added.

Neither error is cured by increasing decimal precision. The wrong mathematical object was computed.

Chapter recap

Exact evaluation needs a complete baseline, support, increment and numerical policy. A successful calculation is not permission to execute, and a closing decomposition is not authorization to credit an owner.

Ownership, capacity and atomic accounting

The most important unimplemented boundary in the new programme is not a missing sum. It is the authority to interpret a particular sum as an owner's capacity change. Software can enforce an adopted rule, but it cannot obtain that rule from the fact that several numbers add correctly.

52.1 A child action is not its owner

A request can be made by a clinic, performed by a carrier and supplied by a depot. The action record may identify all three. None of those labels automatically defines who owns its path attribution. The prospective programme needs an explicit, versioned $\text{owner}(a)$ map.

Once each child has one declared owner, aggregation is straightforward: $\Delta B_i = \sum_{a:\text{owner}(a)=i} R_a$. Closure follows by regrouping the finite sum. The difficult question is not whether the sum can be implemented, but which policy it represents and what claims may be made about it.

52.2 Same-event netting is a policy

Suppose one owner has child lines $+5$ and -3 and starts at zero. Owner-level same-event netting produces $+2$ and passes the proposed affordability rule. A policy requiring each negative child to be prefunded would refuse the negative child. A global-pooling policy could do something else again. These are distinct stochastic processes.

Likewise, a group with owner changes $(+5, -3)$ is not affordable from two zero balances merely because its total is positive. One owner's surplus does not become another's funding under the no-transfer proposal. The implementation must not substitute a single aggregate-balance check for owner-wise checks.

52.3 All-or-nothing effects

If a selected group is accepted under a joint contract, physical and account updates must refer to that same group. A partial credit followed by a failed debit can create spendable capacity without the corresponding authorized

event. A physical update followed by a lost receipt can make replay execute the same action twice.

A later execution design must therefore define commit boundaries, durable identities and recovery behavior across the linked records. “Atomic” here means an all-or-nothing committed event under that contract; it does not mean hiding a multi-carrier physical chain inside one mega-action.

52.4 Audit is not a spendable pool

The external audit J records signed forcing effects. It is not a wallet, loan facility or reward for the next actor. In the cost-free proposal, $V + \sum_i B_i - J = K$. A software routine must be able to check that identity without making J available to the selector as spending capacity.

The invariant alone cannot identify which record is wrong. An erroneous physical update and matching erroneous credit can cancel. Independent conservation, action-identity and owner-level checks are necessary in addition to the aggregate account residual.

52.5 Existing settlement machinery has a narrower meaning

The source-locked framework includes a repaired share-plus-residual closure validator on another lineage [6]. That is an implemented validation mechanism, not an adopted Gaussian owner rule. Its historical amendment’s authoring-stage label is not the whole later implementation history; neither should later implementation be mistaken for scientific approval of the new capacity mechanism.

Chapter recap

Capacity requires a declared ownership and update policy, exact owner-wise affordability, linked event identity and durable all-or-nothing effects. The book specifies the boundary; it does not claim that runtime exists.

Randomness, paired controls and reproducibility

“Use the same seed” sounds like a complete reproducibility instruction. It is not. Two implementations can start from the same seed and consume draws in different orders, causing every later disturbance or choice to diverge. Scientific pairing requires a more explicit account of randomness.

53.1 Separate the sources of randomness

The programme distinguishes nature or forcing draws from actor-selection draws. If proposals are random, their stream also needs an explicit role. A counter-addressed design can associate each draw with coordinates such as stream, epoch, event and attempt, so an extra actor draw does not shift future nature draws. This is a candidate implementation technique, not a substitute for registering the actual stochastic law.

The generator, distribution transform, coordinate schema and retry behavior all belong to the reproducibility contract. Independent seeds alone do not prove independence of every downstream mechanism or prevent a helper from reading a future draw.

53.2 The physical-random comparison

The proposed comparison uses the same declared physical domain and menu construction but omits the EBU-capacity affordability filter. It is not an EBU maximizer, a service optimizer or the historical full multi-edge P1C arm. Those alternatives answer other questions.

Control parity means equal rules where equality was promised, not equal realized actions. Once states differ, physical feasibility can make their menus differ. A comparison must report that difference rather than force one arm to perform an invalid action to keep a superficial pairing.

53.3 Common raw draws versus common applied forcing

Imagine a raw forcing draw proposes moving three units out of a cell. One arm currently holds four there and another holds two. The same raw proposal is feasible for one and infeasible for the other. Rejection, clipping or resampling can yield different applied disturbances.

Thus shared raw randomness does not guarantee shared physical exposure. A future protocol must declare the handling rule, record raw and applied objects separately and determine how exposure differences affect the comparison. Silently promising both forms of equality would be false.

This is not a reason to abandon paired designs. It is a reason to specify what is paired. The scientific question may concern response under common raw environmental opportunities, or response to identical admissible physical inputs. They are different estimands when feasibility depends on the evolving state.

53.4 Restart without a new experiment

A reproducible checkpoint must restore stock, balances, audit, parameter version, pending event identity and random-stream coordinates together. Restoring only the stock can leave an apparently identical physical world with different future affordability or random choices.

Whether a restart is permitted after partial execution belongs to the study's operational contract. A book cannot grant an extra scientific attempt by calling it recovery. Historical Gate 1D-C illustrates why execution attempts and completion sentinels are separately recorded.

Practice question

Two arms have equal initial state and equal nature seed but use one shared random generator for nature and actors. The EBU arm rejects more candidates and consumes extra draws. Are future nature inputs paired? Not necessarily. The rejection path has changed draw consumption. An independently addressed nature stream would remove this particular coupling, while state-dependent forcing feasibility would remain a separate issue.

Chapter recap

Reproducibility concerns the exact stochastic process, not a seed label. Fair comparison requires explicit stream ownership, exposure accounting and restart semantics.

Tests that can genuinely detect being wrong

A useful test is designed so that a plausible defect makes it fail. A suite that merely confirms its own assumptions can stay green while the scientific meaning has changed. The goal is not a large number of checks; it is coverage of the ways the implementation could compute the wrong object.

54.1 Execution classes must be named honestly

Static inspection covers source, schema, hashes and declarations. Pure arithmetic evaluates a fixed mathematical expression without advancing model state. A transition test executes a physical update, even if there is only one synthetic tick. A scientific study executes its registered process and interprets outcomes under a frozen protocol.

These classes have different authorization boundaries. A test's filename or speed does not determine its class. Importing a module can itself have side effects, so calling a test static requires inspection of what it actually does. A source scan that sees no direct transition call cannot prove that an imported helper never transitions.

54.2 Arithmetic checks for the new family

The future Gaussian foundation should compare finite endpoint evaluation with the exact expansion, local with global differences, path sums with group value, and rate/quantity conversions at non-unit durations. It should include reference states, overshooting quantities, cancelling groups and unequal scales, not only beneficial transfers.

An especially useful defect injection is replacing exact finite value by the initial tangent. The quantity-four example changes from twelve to sixteen. Another is summing two same-baseline singleton values: the shared group changes from twelve to fourteen. A negative control should detect those substitutions, not merely run them and count a pass.

54.3 Physical checks are independent

Conservation, nonnegative stock, allowed support and joint capacity must be tested independently of the potential. A negative source and positive destination can preserve total mass. A changed sink can preserve one aggregate residual while violating its declared physical update. Passing one scalar identity is therefore not a complete physical validator.

At the lossless first-domain boundary, any loss coordinate should remain inert under its declared rule. A future loss-aware extension needs its own carrier-by-carrier checks; the success of the lossless suite cannot be reported as certification of that extension.

54.4 Information and account controls

Required prospective controls include forbidden remote reads, future draw access, stale epochs, duplicate settlement, post-selection quantity changes, missing owners, wrong parameter versions and cross-owner pooling. Each should specify the expected refusal and the state that must remain unchanged.

A refused campaign entry is not proven by any nonzero process exit. A syntax error also exits unsuccessfully. The check must verify the intended refusal class and message, and where relevant that no model state changed before it. Refusal is a behavior with a contract, not the absence of a successful printout.

54.5 What green checks do not establish

Even complete conformance to a specification does not prove the specification is scientifically useful. A correctly implemented process can cycle, ration, refuse, or fail to improve its declared metric. Those are possible scientific outcomes, not automatic reasons to weaken tests or alter the world.

This book's worked arithmetic is checked independently as teaching material. Those checks are not reported as implementation acceptance counts. Historical Q-test and D-test counts in Appendix H-III remain attached to their own code and configurations; none is inherited by an unbuilt Gaussian runtime.

Chapter recap

Test the equation, the physical transformation, information boundaries, accounting semantics and execution guards separately. Then keep conformance distinct from scientific success.

Provenance, execution and immutable evidence

A result is not reproducible merely because its final number is saved. Another reader needs to know which model, protocol, implementation and environment produced it, whether the run completed, and whether the records were changed afterward. Provenance connects a claim to that chain.

55.1 The scientific packet precedes the run

A registered packet identifies hypotheses, domain, arms, parameters, randomness, metrics, falsifiers and interpretation rules. It also binds the implementation and execution environment. These decisions must precede outcome inspection; otherwise a successful-looking configuration can be selected from many attempts without saying so.

The present book does not fill those fields for the Gaussian programme. It describes what must be declared. No horizon, seed family, numerical tolerance, performance threshold or cloud resource is frozen by an example in these chapters.

55.2 Attempt, run and completion are different objects

An invocation can fail before a scientific state transition. A campaign can contain many registered runs within one authorized attempt. A runner can finish producing output before the full study is durably finalized. These distinctions prevent both inflated execution claims and unauthorized retry disguised as routine continuation.

Historical Gate 1D-C's manifest explicitly distinguishes the runner summary from the full-study completion sentinel [8]. That exact operational contract belongs to that study. A future Gaussian packet must define its own compatible mechanics and permissions, not inherit an old launch command.

55.3 Content identity and scientific meaning

Hashes identify bytes. A manifest should record the protocol, configuration, source versions and produced artifacts so their identities can be checked. Hashes do not certify a model's validity or a sensor's honesty. They make it harder to substitute one object for another unnoticed.

Canonical serialization matters when equivalent-looking objects can produce different byte sequences. Numeric representation, key ordering and schema version are part of the identity contract. A replay should not silently normalize a value in a way that changes the registered scientific object.

55.4 What to retain in a trace

For each event, retain enough information to reconstruct its physical baseline, forcing, candidate/selected identities, accepted increment, valuation inputs, owner changes, account audit and refusal state. The exact schema is prospective, but its purpose is fixed: an independent reader must be able to tell what happened without trusting a narrative summary alone.

Not every diagnostic belongs in the actor's view. A global evaluator may record quantities for analysis while the selector is forbidden to read them. Trace completeness and actor information locality are compatible when their data paths are separated.

55.5 Corrections should not erase history

If a result artifact is wrong, the correction stage should identify the original, the defect, the corrected interpretation and the authority for any replacement. Rewriting the original bytes until they fit a later story destroys the very evidence needed to understand the error.

The book revision follows the same principle. Original PDFs and historical result artifacts are preserved. New exposition explains which model-specific claims do not transfer, rather than editing old experimental outcomes to look like Gaussian evidence.

Chapter recap

Provenance makes a claim inspectable, not automatically true. Frozen scientific choices, exact identities, completion rules and preserved failures allow an independent review to determine what the evidence actually supports.

The historical engine and the Onsager connection

The older implementation is not simply the new model with a different potential expression. It has a particular natural drive, flux controller, permission layer and event order. Understanding those choices is necessary both to preserve its results and to avoid importing them accidentally.

56.1 Why force, mobility and flux entered EBU

Onsager’s primary papers concern coupled irreversible processes and linear force–rate relations under microscopic-reversibility assumptions [1, 2]. The useful modelling pattern for EBU was to separate a directional driving force from a response coefficient. The earlier D0 design made that relationship explicit instead of treating the potential’s derivative as though it were already a physical flow.

Its local marginal came from the historical hinge potential. For an edge with usable-delivery efficiency η_e , the directional force was $f_e = \mu_i - \eta_e \mu_j$. The controller requested

$$J_e = M_e[f_e - \theta_e]_+. \quad (56.1)$$

Mobility M_e controls responsiveness; θ_e is an edge activation threshold. It is distinct from Allee or homeostatic stock thresholds. Neither coefficient is an EBU balance or a source’s physical stock.

This adaptation is called Onsager-type because the model declares its own potential and directed threshold response. It does not establish that the potential is thermodynamic free energy, or that a social network satisfies microscopic reversibility or reciprocal transport coefficients. The reason to use it is the transparent conditional force–flux calculation, not the prestige of an analogy.

56.2 D0’s synchronous update

The historical map freezes state, computes the natural drive $u(x)$ and edge requests, then applies a synchronous Euler update of the form $x^+ = x + \Delta t(u +$

SJ). Frozen synchronous evaluation prevents a newly updated neighbor from influencing another edge within the same nominal step. Sequentially mutating an edge loop would implement a different propagation rule.

Part II derives the continuous-time dissipation structure and its finite-step curvature correction. The historical tests concern their registered hinge model and certificates. Replacing the evaluator with a Gaussian formula while retaining the old drive and selector would create a hybrid model, not automatically implement the new programme.

56.3 P1C is a permission layer with premises

The historical P1C classifier distinguishes regenerative and other source types and their admissibility states. The robust budget's arithmetic assumes the applicable source classification. For a finite or irreversible stock, the preservation policy does not become an internal-transfer permission merely because a helper can compute a positive number.

The distinction between R_{eff} and homeostatic R also survives. The former bounds export under a source certificate; it does not itself deliver resource to a destination below its own reference or reserve. A resolver's protection cannot be used to delete another controller's service logic while claiming the same function remains.

56.4 What is retained and what is not imported

Keep historical D0/P1C, quote code and evidence immutable. Reuse reviewed ideas such as complete support, frozen identities and exact finite evaluation. Do not import Allee drive, gradient-best selection, service-first comparators or finite-source export prohibitions into the Gaussian random actor process without a separate scientific declaration.

Likewise, retaining the Onsager explanation does not make it the active actor policy. It remains a mathematically meaningful controller alternative and an important part of the project's history. The new process asks a different question about random affordable actions.

Chapter recap

The historical engine supplies mechanisms, proofs and evidence at a known scope. Its force–mobility–flux architecture should be explained, not hidden; its controller should not be silently reused under a different study name.

What the historical evidence actually establishes

A research programme becomes more credible when it preserves the conditions under which it failed or produced a flat result. Replacing every older chapter with a success-oriented summary would lose that information. This edition therefore retains selected original evidence chapters as historical source pages and adds a careful reading guide.

57.1 Read a result together with its model

Appendix H-III reproduces original Chapters 53–59, covering D1–D10, Gate 1B, adversarial/service-alignment work and O14. The old potential, regeneration, preservation rules and selectors remain attached to those accounts. Their numerical results were not recomputed under the new formula.

The appropriate question is not “Did EBU pass?” without qualification. Ask which equation, physical domain, comparison, horizon and falsifier were tested. A test of quote-code conformance and a test of service alignment answer different questions even when both involve the same numerical quote.

57.2 Why the old distinctions still teach useful lessons

The original soft-versus-hard preservation comparisons distinguish a penalty from a permission boundary. That lesson survives a change in potential: assigning a large burden does not physically prevent an action. The observational-quote comparison distinguishes logging from influencing the plant. The adversarial examples ask whether a locally closing account can still coexist with unrepresented or persistent harm.

These are methodological lessons. They are not evidence that the Gaussian capacity policy has passed analogous tests. A new actor, menu or state representation changes the tested system and requires its own registered evaluation.

57.3 O14 and a later finalized gate are different records

The original O14 chapter preserves its own capability and flat-outcome analysis. It should not be relabelled as Gate 1D-C or as Gaussian evidence. Some older analytical documents also retain operational status text from before the later gate was completed. Their mathematical content can be useful without treating their historical status paragraphs as current.

The later committed Gate 1D-C manifest records a finalized study [8]. Its execution SHA is `8793bda`; its inventory contains thirty registered runs and six thousand trace rows. The recorded stdout specifies three worlds, two timestep settings and five arms, with two hundred ticks per run. Its reported outcome counts are twenty `physical_impossibility`, eight `safe_rationing_physical_scarcity`, and two `preserve_and_serve`; it reports no fired falsifier.

Those are narrow historical source facts, not thirty independent proofs of the accounting theory or a long-run Gaussian result. The revised books do not rerun, tune, extend or reinterpret the frozen campaign. The source manifest and original scientific protocol govern its actual claims.

57.4 A comparison can be invalid even when its numbers are real

An arm allowed several simultaneous deliveries has a different capability set from an arm restricted to one action per configured source. Such an arm can be informative as a capability benchmark, but it is not automatically a matched test of the contribution of the EBU selector. Historical comparison restrictions must remain visible when reading an outcome table.

Likewise, a new Gaussian arm with a different physical menu cannot claim an advantage of capacity accounting alone unless that difference is part of the registered estimand. The artifact may be genuine while the proposed cross-model inference is unsupported.

57.5 What preservation of the appendix means

The appendix is reproduced source material, not a fresh independent audit of every archived run. Its figures and statements keep their original scope and publication-era wording. The new source manifest identifies the exact PDF pages. Readers can return to the full preserved originals for cross-references outside the extract.

This treatment avoids two opposite errors: discarding useful history because the active model changed, and making the new model appear validated by repainting that history. Neither is necessary for a coherent series.

Chapter recap

Historical experiments remain evidence about their own registered systems. They guide questions and expose pitfalls for new research, but they cannot answer an unexecuted Gaussian comparison by analogy alone.

Measurement, calibration and the loss boundary

A perfectly implemented equation can return a misleading number if its inputs do not represent the physical event. This is why a book about code must return to measurement: what quantity changed, where was it measured, what uncertainty remains, and who justified the reference against which the change is evaluated?

58.1 References and scales need an interpretation

The Gaussian reference x_i^* describes a declared functional condition. The scale σ_i describes how deviations are weighted. Neither is automatically an observed mean, a measurement uncertainty or a just allocation. A physical unit conversion rescales stock, reference and scale together; changing only the scale is a substantive change in valuation.

An empirical programme would need a parameter record explaining the functional evidence, measurement domain, validity period and uncertainty of each declaration. That work is not supplied by choosing convenient numbers for a textbook example. The examples here are explicitly illustrative.

58.2 Measurement and account boundaries

Source withdrawal, useful destination delivery and loss are different measurements. A pipeline meter at the source does not establish delivery at the destination. A receipt should retain those distinctions even in a first lossless model, where the declared values coincide.

Physical time matters too. An endpoint observation does not necessarily identify the intermediate path of several simultaneous actions. If the account uses endpoint interpolation, record that convention. If it claims a model-realized constant-rate path, identify the physical assumption and the evidence required to use that reading outside the model.

58.3 Conservation does not choose a valuation

An explicit loss coordinate can close total physical accounting. It does not automatically become an argument of the homeostatic potential, and excluding it from that potential does not automatically authorize an additional charge. The represented consequence, process burden and physical carrier must be reconciled without double counting.

The historical Definition 6.4 discusses a dissipated-stock charge with a rationale that no destroyed-stock coordinate is present [4]. Later candidate work includes a loss ledger coordinate. The statements “absent from state” and “present in a ledger but absent from V ” are not literally identical. A broader loss-aware rule needs an explicit authority/wording reconciliation; this book does not silently choose one.

The active first domain has $\eta = 1$ and $C = 0$. Loss-aware runtime is not certified by that choice. An aggregate conservation residual can hide an invalid sink update if the declared conservation profile is not enforced in the actual transition. Carrier-specific validation remains necessary.

58.4 Do not invent burdens to complete a story

Fuel conversion, electricity delivery, pumping and water transfer can be represented as linked atomic transformations when they are in scope. Their existence is not a reason to add an arbitrary extra cost to every downstream action. Each physical effect needs a declared carrier and one justified valuation location. The chain must neither disappear inside a mega-action nor be charged repeatedly under several names.

Monetary price, inconvenience, sanctions and physical process burden also have different meanings. A payment may explain how an essential service is financed without becoming a physical term in the EBU equation.

58.5 Governance and privacy are not residual variables

Selecting references, identifying owners and deciding who may access measurements have institutional consequences. A model that calls itself local does not automatically settle consent, privacy, auditability or access. Those questions require an explicit governance layer and appropriate domain review; they cannot be solved by adding a coefficient to V .

The practical benefit of keeping records distinct is transparency. A necessary action can have negative represented field value without the account declaring that its recipient should be denied service. The limited experimental capacity

policy must not be presented as a ready rule for real-world access to essential resources.

Chapter recap

Measurement and calibration determine what a precise calculation means. The first lossless world allows focused mathematics while leaving, rather than pretending to solve, the broader loss and institutional boundaries.

What a long-run study could show

The future study should be capable of finding that the proposed mechanism does not help. If every unfavorable outcome is reclassified as an invalid world or repaired by changing the actor, the programme ceases to test its own question. Interpretation must therefore be designed before the outcomes are visible.

59.1 A single disturbance and continuing forcing

The roadmap distinguishes a response after one declared disturbance from a continuously driven stochastic regime [7]. After one shock, forcing stops and the external audit remains fixed. The physical potential and total capacity can exchange subject to the conditional invariant. The process may recover, cycle, become trapped or refuse all available actions.

Continuing forcing changes the question. Even if physical deviation appears to stabilize, external audit and accumulated capacity may drift. A claim of stationarity must name its object: the physical state, potential, selected observables, balance distribution, or the entire augmented process. A flat-looking plot of one coordinate does not establish stationarity of all.

59.2 Boundedness is not the benefit metric

Finite conserved mass and positive fixed scales already bound the Gaussian potential on its domain. The physical-random control has that same bound. Therefore “the EBU arm did not diverge to infinity” is not a discriminating success claim for this first closed model.

Potential occupancy, time away from reference, recovery after disturbance, refusal, cycling and distribution are possible families of observables. They are not frozen metrics here. Their definitions, aggregation, uncertainty treatment and falsifiers belong to a later preregistration. Service belongs in the interpretation only if the plant actually defines and measures it.

59.3 Physical validity precedes performance

A trajectory that violates conservation or declared physical constraints cannot support a performance claim merely because its potential is low. Similarly, an invalid comparator undermines a comparative claim differently from a valid comparator that misses its own target.

It is helpful to retain raw facts separately: physical validity, each arm's measured outcomes, and any oracle or feasibility verdict. Derived labels should not force unresolved cases into success or failure by default. If a claim requires an oracle that was not resolved, report that inference as inconclusive while preserving the observed facts.

59.4 No-action is a scientifically meaningful possibility

At reference with zero genesis, the proposed account blocks nonzero-net negative-valued departures. After a disturbance, however, menu or ownership structure may still leave no affordable action. A refusal can therefore be an intended consequence of the declared rules, a reachability limitation, or a specification defect. The records must distinguish them.

Adding a free initial balance, refill, loan or target-directed rescue to avoid refusal would change the experiment. Such alternatives can be studied later, but must not be inserted after seeing an inconvenient result and reported as though the original policy succeeded.

59.5 From finite evidence to broader claims

A finite registered study can support statements about its tested regimes and uncertainty. It does not establish unlimited-horizon safety or economic validity. A theorem may establish an invariant for all permitted events, yet leave the quality of the resulting dynamics unresolved. Evidence and proof can complement one another without substituting for one another.

The strongest report includes adverse and null comparisons, actual exposure, software violations, missing data and protocol deviations. It distinguishes what was observed from the causal explanation offered for it. Where the design cannot separate explanations, the report should say so.

Chapter recap

The long-run question remains open because accounting identities constrain but do not choose behavior. An honest study needs registered observations that can distinguish mechanisms, not a success label guaranteed by geometry.

A worked audit and the next research boundary

We end by reviewing a complete illustrative record. The point is not to execute the future engine on paper. It is to identify exactly which facts a reader would need in order to check one event, and which conclusions remain unavailable even after the arithmetic is correct.

60.1 The declaration

Use two cells, reference $(10, 10)$ and scales $(2, 2)$. Begin at reference with zero capacity and zero external audit. A declared conservative forcing increment $(4, -4)$ creates baseline $(14, 6)$. Its potential is four. Two accepted child actions each transfer one unit from the first cell to the second. The aggregate increment is $(-2, 2)$ and endpoint $(12, 8)$.

This is an illustrative declaration, not a registered world. It does not select a random law, timestep, horizon or performance target. The children are fixed mathematical objects for the audit.

60.2 The physical and field checks

Total stock is twenty in every listed state and every stock is nonnegative. The accepted source withdrawal is two, within its fourteen-unit baseline. If there were another declared physical constraint, it would also need checking; the absence of such a field from this example is not permission to ignore it in another model.

At baseline the marginals are $(1, -1)$. Each child has increment $(-1, 1)$, and the aggregate curvature term is $\frac{1}{2}(-2, 2)^T H(-2, 2) = 1$. The tangent group benefit is four, so the exact group value is three. Endpoint evaluation agrees: $4 - 1 = 3$.

For the straight common path, each child attribution is $3/2$. Two standalone singleton values would instead be $7/4$ each and sum to $7/2$, overcounting the group by one-half. This discrepancy is not a floating-point artifact; it is the exact shared-field cross term.

60.3 The account check and its missing authority

Suppose, conditionally, the adopted policy assigns both children to one owner. That owner's capacity change would be three. If each child belongs to a separate owner, each would receive $3/2$ under the same declared path-to-capacity policy. Both pass zero-genesis affordability in this example. Neither assignment is derived from the sum identity; the owner rule is a separate input that the actual programme still must adopt.

The external audit is four. With total capacity three and endpoint potential one, $V + \sum B - J = 1 + 3 - 4 = 0$. This verifies the conditional aggregate account. It does not verify the provenance of the owner declaration, the sensor readings or the random selection that supposedly chose the group.

60.4 A list of claims the record cannot support

It cannot show that the action was sampled uniformly, because no random stream or menu record has been supplied. It cannot show that the process will keep recovering, because only one fixed event is described. It cannot show fairness, because arithmetic closure does not define fairness. It cannot show causal entitlement to the child amounts, because the path decomposition is not an intervention study.

It also cannot establish that a runnable Gaussian engine exists. A future implementation must faithfully create, execute and preserve these objects under its accepted profile, not merely print the same numbers for one example.

60.5 The claim ledger

Claim	Current status in this edition
Finite Gaussian expansion	Derived exactly under fixed quadratic geometry.
Local affected-factor cancellation	General conditional identity.
Common-path sum	Exact decomposition; no ownership conclusion.
Onsager-type dissipation	Conditional on the specified flux law and domain.
Capacity invariant	Conditional on the proposed signed owner/update policy.
Gaussian runtime acceptance	Not established by this book.
Gaussian scientific outcomes	No new study executed.
Historical Gate 1D-C completion	Recorded by its separate committed manifest.

60.6 The next separately authorized stages

First resolve the load-bearing event, owner/path, menu/locality, forcing coupling and numerical-policy contracts. Then implement reviewed generic reuse and isolated Gaussian evaluation. Validate pure foundation properties before separately authorized transition fixtures. Only after that should a scientific packet freeze the study configuration, comparators and interpretation rules. Execution requires packet-specific permission.

This sequence does not require the author to decide every routine coding detail. It identifies the decisions that change the scientific object. Routine implementation can proceed within an accepted contract; it must stop where a missing choice would create a different process.

Stage A and Stage B are separate study questions, not automatic continuation flags. A single-shock outcome, including nonrecovery, needs a disposition before a continuously forced regime is treated as the next approved study. Neither stage is automatically SD-01, and no AWS launch or cost is authorized by a manuscript explaining the programme.

60.7 What this revision accomplished

The three books now have a consistent active Gaussian vocabulary and a visible connection to the original force–mobility–flux mathematics. The historical

record remains identifiable rather than being rewritten as new evidence. The derivations and worked arithmetic can be reviewed now while implementation and empirical claims remain at their actual status.

That is not a delay in doing science. It is the distinction that lets the next experiment answer a clear question. A model can be implemented, tested and found wanting without destroying the exact identities it was built to respect. Conversely, correct identities do not guarantee the model deserves institutional adoption. Both possibilities must remain open.

Sources and authority map

- [1] Lars Onsager. *Reciprocal Relations in Irreversible Processes. I*. Physical Review 37, 405–426 (1931). doi:10.1103/PhysRev.37.405. Historical primary source; no economic application is inferred.
- [2] Lars Onsager. *Reciprocal Relations in Irreversible Processes. II*. Physical Review 38, 2265–2279 (1931). doi:10.1103/PhysRev.38.2265. Linear response, microscopic reversibility and dissipation-function context. The EBU threshold-mobility controller is a separately declared adaptation.
- [3] Energy Balance Project. *Local Gaussian EBU: Programme Reconciliation*. Committed source LOCAL_GAUSSIAN_EBU_PROGRAMME_RECONCILIATION.md, inspected at 05f3cb0. Sections 3–8 supply the prospective domain and conditional accounting identities; Section 10 retains the open decisions. Not an executed study.
- [4] Energy Balance Project. *V3.0 Local EBU Foundation Draft*. V3.0_LOCAL_EBU_FOUNDATION_DRAFT.md. Sections 6–8: exact finite quotation, cost categories, local cancellation and scoped sequential identities. Historical D0/P1C assumptions remain attached to their statements.
- [5] Energy Balance Project. *The Sequential–Parallel Bridge of EBU*, analytical checkpoint v0.2. SEQUENTIAL_PARALLEL_BRIDGE.md, Sections 3–7. Joint state changes, telescoping and comparator-relative interaction. Its operational status passages are historical, not a current Gate 1D-C completion record.
- [6] Energy Balance Project. *Local Gaussian Programme: Framework Decision*. LOCAL_GAUSSIAN_EBU_FRAMEWORK_DECISION.md. Source-locked F2 reuse decision for planning, with separate event, ownership, capacity and locality dependencies. Does not authorize scientific runtime.
- [7] Energy Balance Project. *Local Gaussian EBU: Deferred Implementation Roadmap*. LOCAL_GAUSSIAN_EBU_IMPLEMENTATION_ROADMAP.md. Implementation, transition fixtures, preregistration and execution are separate stages. Historical book-stage wording is superseded only by the author’s later book-revision instruction.

- [8] Energy Balance Project. *Gate 1D-C Finalized Execution Manifest*, `results/v3.0/gate1dc/MANIFEST.md`; accompanying registered protocol, JSON plan, summary and stdout. Historical finalized evidence: 30 runs and 6,000 trace rows. Execution identity 8793bda. No Gaussian inference is made.
- [9] Konrad Grzyb. *The Energy Balance Project: Unified Explanatory Edition*, Parts II and III. Author-supplied PDFs, 160 and 153 pages. Content/style baselines and sources of the explicitly marked historical appendices. Original artifacts remain unchanged.
- [10] Konrad Grzyb. *Physical Intuition and the EBU Idea*, revised Gaussian explanatory Part I. Author-review edition. The shared Ridge–Vale examples and typographic language continue here; examples are mathematical teaching constructions, not experimental evidence.

How to verify a source claim

The companion source manifest records full file hashes, the checkout identity, the original PDF identities and the exact original pages retained in each appendix. The outline disposition records subjects rewritten rather than claiming every original sentence survives. All new derivations state their assumptions in the chapter itself. A citation identifies a source; it does not transform a prospective design into an accepted theorem or experiment.

Historical Appendix H-III

The following forty-five pages preserve original Part III Chapters 53–59, original PDF pages 66–110. They contain the original accounts of D1–D10, Gate 1B, adversarial/service-alignment work and O14. Original equations, outcomes, qualifications and folios remain; every page gains a historical margin banner, and chapter titles are displayed smaller. This is a source-preserving archive, not new experiments.

The old potential contains lower/upper/reserve hinge terms. Its natural drive, P1C permission and selector choices are model-specific. They have not been replaced inside these records by Gaussian formulas. Nulls, failures and limitations are as important to preserve as favorable findings.

Publication-era words such as “current” remain historical. Chapter 57 separately identifies the later finalized Gate 1D-C source and its narrow scope. Original cross-references lead to the preserved original volume; they are not revised-edition chapter references. Source reproduction is not a fresh independent reanalysis of every archived study.

The first wind tunnel: D1 through D4

Before testing preservation or actor accounting, the project placed the D0 engine in small deterministic fixtures. A wind tunnel does not reproduce the entire atmosphere; it isolates forces that would be difficult to interpret in a complete aircraft flight. In the same way, D1 through D4 isolate descent, transport loss, synchronous causality, regeneration, and service.

The policy labels used in this sequence are:

- P0: no transport;
- P1: the exact synchronous loss-aware D0 law;
- P1K: a diagnostic projection wrapper, outside the theorem;
- P2: a deliberately loss-blind force;
- P3: a deliberately sequential live-state update;
- P4: a deliberately oversized timestep.

The controls are not competing economic systems. Each removes or changes one specific property so that the experiment can detect its importance.

53.1 D1: can the smallest world move downhill?

D1 contains two cells, no external drive, and a lossless edge. The initial state is $(19, 2)$ and the initial burden is 17. Under P0 nothing moves, so the final burden remains 17. Under P1 with $\Delta t = 0.5$ and certified ratio $r_{\Delta t} = 0.5$, the field relaxes until the final burden is approximately 6.31×10^{-30} , numerically zero. There are no descent violations.

The result is simple by design. It checks that a clearly active edge sends stock in the direction predicted by the local field and that the discrete update agrees

with the expected undriven relaxation. It does not test regeneration, scarcity, service, or reserve safety.

How to read the numbers

“Final burden near zero” means both cells ended in a state where the represented penalty was essentially absent. It does not mean the stocks became zero. In fact, the observed stocks remained between 4.5 and 16.5. Burden and stock are different quantities.

53.2 D2: why route efficiency changes the correct resting point

D2 uses a lossy edge. P1 and the loss-blind P2 begin from the same state (8.5, 2) with burden 66.25. The difference is one factor: P1 uses $f = \mu_i - \eta\mu_j$, while P2 behaves as though $\eta = 1$ inside the force.

P1 ends with burden 58.243 and zero descent violations. Its final loss-aware force is approximately 0.05, equal to the activation threshold, so the edge rests. At the same state the loss-blind force is still about 5.294. P2 therefore continues moving material after the correct loss-aware law has reached its resting condition. It ends with the higher burden 61.814 and records eleven descent violations.

This fixture turns an abstract efficiency term into a physical story. Imagine a clinic reachable only through a route that destroys half the medicine. The benefit of one unit leaving the depot is not the benefit of one unit arriving. Ignoring efficiency can make continued transfer look attractive after its actual delivered benefit no longer justifies source withdrawal.

What the negative control proves

P2 shows that the fixture and diagnostics can detect omission of η . It does not prove that every real efficiency is known exactly. Measurement uncertainty is a later problem.

53.3 D3: one-hop causality

D3 places three cells in a chain $0 \rightarrow 1 \rightarrow 2$. It runs paired baseline and perturbed initial states under both synchronous P1 and sequential P3. The measured quantity is not one absolute cell value. It is the difference at the probe cell 2 between the paired runs.

Under synchronous P1, the perturbation produces zero probe difference at tick 1 and first appears at tick 2. Under sequential P3, the probe differs by 0.0162 at tick 1. The sequential loop updated the first edge and then allowed the second edge to read that newly changed relay state inside the same nominal tick.

This may sound like a small programming detail, but it changes the model's notion of time. In P1, one tick represents one simultaneous response to a common observation. In P3, stored edge order determines how far information can travel. A route listed earlier in a file can gain causal priority without any physical meaning.

53.4 D4: drive, regeneration, and useful service

D4 introduces external dynamics and demand. P0 serves only 6 of 150 requested units before the consumer leaves the declared physical domain; its minimum state reaches -144 . P1 serves all 150 requested units, remains physically admissible, and ends with burden 4.578. P1K is identical here because no projection is needed.

The burden does not decrease on every driven tick. P1 records many “descent violations” relative to an undriven monotonic expectation because demand and regeneration continuously inject and remove stock. This is not a contradiction of the undriven theorem. The correct driven inequality contains the term $\mu^T u$; natural drive can raise the field burden while transport still acts dissipatively.

Service is counted only while physically valid

The P0 destination continues to subtract demand after crossing below zero in the unconstrained equation. The study does not count that mathematical subtraction as valid delivered service. Its service total stops when the physical domain is lost. This prevents an impossible negative store from satisfying an

accounting target.

53.5 A trace-reading example

To read one D4 tick, follow this sequence:

1. inspect x^n and verify that all cells are within the declared domain;
2. read the local drive vector u ;
3. inspect the edge force and flux calculated from the frozen views;
4. multiply each rate by Δt to obtain stock changes;
5. confirm the source withdrawal, delivered amount, and transport loss;
6. inspect x^{n+1} and the ledger residual;
7. only then read burden and service diagnostics.

This order prevents a common interpretation error: starting with a final score and guessing backward which physical event produced it.

53.6 What D1–D4 established

These fixtures support the following finite implementation statements: the released P1 engine relaxed the simple lossless world; loss awareness changed the resting state and avoided the P2 violations; synchronous evaluation enforced one-hop causality; and the driven P1 world served the registered demand while remaining in domain. They also demonstrate that a no-transport baseline can be physically invalid and that undriven descent is not the correct success criterion for every driven tick.

They do not establish hard reserve preservation. That question appears only when a source can cross a dangerous regenerative boundary.

Chapter recap

D1 checks simple downhill relaxation. D2 makes efficiency observable. D3 makes synchronous causality observable. D4 adds drive and service and shows why burden, domain validity, and useful delivery must be read separately.

Together they test the physical substrate before the preservation controller is introduced.

Exercises

1. Why is P0's constant burden in D1 not evidence that P0 is a good policy?
2. Describe a real scheduling system in which sequential order would create an artificial priority similar to P3.
3. D4's P1 burden is not monotone on every driven tick. Explain why that does not invalidate the undriven descent theorem.

Learning from D5 through D8

The next four fixtures are valuable because they did not produce a simple series of victories. D5 failed to discriminate a reserve penalty. D6 and D7 showed recovery under shocks but did not create a hard preservation result. D8 confirmed one oversized-step failure while falsifying a stronger expectation about repeated growth. Keeping these outcomes made the later protocol better.

54.1 D5: a reserve fixture that was too forgiving

D5 compared the field with reserve weight $\chi = 1$ and $\chi = 0$. The intended question was whether the soft regenerative-danger term would protect an Allee-like source. Both configurations served all 150 requested units, remained in the declared domain, and recorded no reserve or Allee crossing. Their headline trajectories were effectively non-discriminating.

The correct conclusion is not “the reserve term is unnecessary.” The source and demand combination never forced the policies into a region where their difference mattered. A fire alarm cannot be validated in a building where no smoke reaches either sensor.

This distinction became a design rule: a study of a protection mechanism should include a fixture in which the relevant boundary is reachable under the control policy. D9 was later built to satisfy that requirement.

54.2 Why a null result can be useful

A non-discriminating world reveals weakness in the measurement instrument. It can show that initial stock is too large, demand too low, duration too short, or an outcome metric saturated. The scientific response is to preserve the null, diagnose it, and preregister a new discriminating study – not to change the old world until the desired result appears.

This lesson returns in O14, where genuine policy differences again fail to change the registered primary service outcome.

54.3 D6: a supply shock and a viability gap

D6 runs for 1000 ticks and imposes a registered supply shock. P0 leaves the physical domain, serves only 6 of 200 requested units, and does not recover. P1 serves all 200, remains in domain, and satisfies the registered recovery criterion at tick 440. Its final burden is about 1.031.

The recovery metric uses a pre-shock reference window, a threshold relative to that reference, and a sustained post-shock window. This is stronger than calling the first upward tick “recovery.” A brief rebound can occur inside an overall decline.

The fixture also reveals a *viability gap*: useful service and field burden can recover at different rates. A system may continue meeting current demand while some cells remain outside their preferred bands. One summary number cannot tell both stories.

54.4 D7: a demand shock

D7 increases demand during the run. P0 again exits the domain and validly serves only 6 of 260 requested units. P1 serves all 260, remains physically admissible, and reaches its sustained recovery condition at tick 611.

Supply and demand shocks have different local meanings. A supply shock reduces the source’s drive. A demand shock increases the destination’s draw. Both can produce a growing deficit, but they suggest different interventions in a real needs system: diversify supply in one case, improve efficiency or capacity in the other.

The model’s ability to distinguish those mechanisms depends on receipts that keep drive, transport, and service separate.

54.5 D8: deliberately crossing the timestep certificate

Discrete descent requires a sufficiently small timestep. D8 compares an unsafe P4 run with $r_{\Delta t} = 1.2$ to a certified P1 contrast with $r_{\Delta t} = 0.8$. The unsafe

run begins at burden 2 and jumps to 3.92 on the first tick, a ratio of 1.96. The contrast falls to 0.72 without a descent violation.

The unsafe result demonstrates overshoot. A large explicit step can cross the valley of a convex penalty and land higher on the other side, even though its initial force points downhill.

The preregistration also expected a stronger monotone-growth pattern after the first overshoot. That sub-claim failed: the P4 burden did not keep increasing on every subsequent tick. The correct report therefore contains both facts – the immediate counterexample succeeded, while the stronger trajectory description did not.

54.6 The dimensionless timestep ratio

The ratio

$$r_{\Delta t} = \frac{\Delta t}{\Delta t_{\text{certificate}}} \quad (54.1)$$

compares the chosen tick to a computed certificate. Values at or below 1 are inside the declared bound; values above 1 are outside it. Because the ratio is dimensionless, it can compare worlds with different time units and curvature scales.

Being inside the certificate is sufficient for the stated finite-step inequality under its assumptions, not necessary for every trajectory to behave well. Being outside does not guarantee failure on every state. D8 supplies a concrete state where failure is actually observed.

54.7 What changed after these fixtures

Three methodological decisions follow from D5–D8:

1. Protection studies require a boundary-capable positive control.
2. Shock recovery requires persistence, not a single favourable tick.
3. Timestep claims must report both certificate status and actual numerical behaviour.

Most importantly, the project stopped treating a stronger soft penalty as a substitute for a hard source-level guarantee. D5 did not show the soft term

was effective; it showed that the fixture could not answer the question. The next study therefore made the reserve cross under the reserve-blind arms and asked whether an algebraic cap prevented it.

Chapter recap

D5 preserved a non-discriminating null and motivated a harder world. D6 and D7 showed finite shock recovery under P1 while exposing the failure of the no-transport baseline. D8 made timestep overshoot visible and preserved a failed sub-claim. These are examples of experiments improving the research question, not merely confirming a model.

Exercises

1. Why might two reserve-weight settings fail to distinguish the policies?
2. Why should a recovery definition include a sustained window?
3. Explain the difference between “outside the certificate” and “observed to fail.”

D9: when a soft warning meets a hard boundary

D9 is the first preservation fixture designed so that the competing mechanisms must reveal themselves. An Allee-like source begins above its certified reserve, serves sustained demand, and can cross both the reserve and the lower regenerative threshold. The study compares a reserve-blind law, a soft reserve penalty, and the hard P1C cap.

55.1 The world and why it discriminates

The source begins at $x_0 = 13$. Its certified reserve is $R^{\text{eff}} = 11$, the Allee threshold is $A = 5$, the timestep is 0.2, and demand is 5 stock units per unit time for 200 ticks. The world is not made safe by an enormous initial buffer. If the controller continues serving without a hard cap, the source reaches the dangerous region during the registered horizon.

There are four arms:

- D9-A: reserve-blind P1, $\chi = 0$, no hard cap;
- D9-B: soft reserve penalty, $\chi = 1$, no hard cap;
- D9-C: P1C hard cap with $\chi = 1$;
- D9-D: P1C hard cap with $\chi = 0$.

The C-versus-D pair isolates the hard cap from the soft field term. If they are identical, the cap rather than χ determines the boundary behaviour in this fixture.

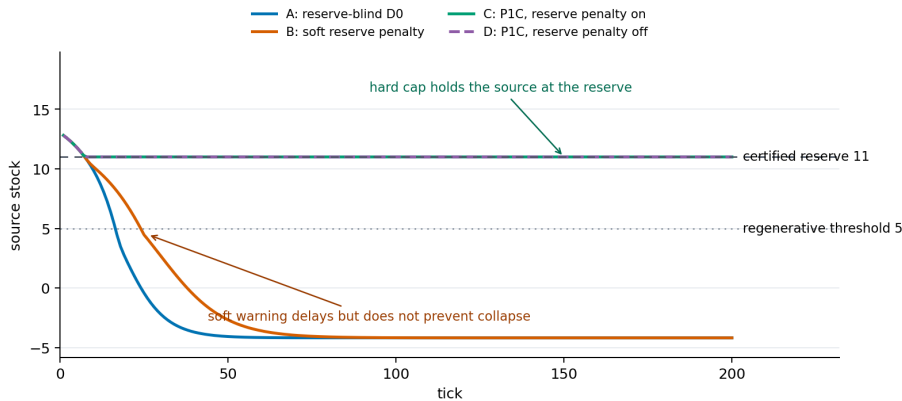


Figure 55.1: *D9 source-stock trajectories generated from the committed per-tick trace. Horizontal lines show the certified reserve and Allee threshold.*

55.2 How to read the trajectory figure

Read Figure 55.1 from left to right. All four arms begin at the same source stock. The blue reserve-blind and orange soft-penalty trajectories both cross the reserve at tick 8. The soft term changes the later path: it delays the Allee crossing from tick 17 to tick 24 and reduces cumulative delivery. It does not create invariance.

The green and dashed purple P1C trajectories lie on top of one another. They descend to 11 and remain there. P1C binds on 193 of 200 ticks. The overlap is evidence that, once the hard constraint controls quantity, switching the soft weight on or off does not change any recorded headline metric in this world.

55.3 The exact committed outcomes

Arm	Reserve tick	Allee tick	Delivered	Unmet
A	8	17	189.692	10.308
B	8	24	172.831	27.169
C	never	never	130.825	69.175
D	never	never	130.825	69.175

The reserve-blind arm delivers the most during the finite horizon, but it does so while entering the collapse basin. The soft arm sacrifices some delivery

and delays deeper damage, yet still crosses both boundaries. P1C delivers less and records the remaining need as unmet demand.

Calling only one column “success” would destroy the meaning of the study. Delivery without preservation consumes tomorrow’s source. Preservation without reporting unmet need can conceal human harm. D9 therefore classifies C and D as *safe rationing*, not as unqualified success.

55.4 A tick at the reserve

Once the P1C source reaches 11, imagine its local drive during a tick adds exactly the quantity that can be safely exported. The robust budget accepts that rate, and the successor returns to 11. If requested export is higher, proportional scaling reduces it. This repeated equality explains the flat green and purple lines: the controller is not freezing all activity; it is allowing only the flow compatible with the boundary.

The observed theorem-eligible count is 200 and the observed violation count is zero for both P1C arms. That is implementation conformance at these ticks. The algebraic theorem remains the reason the inequality should hold under its assumptions.

55.5 Comparing tick 7, tick 8, and the later plateau

The first reserve crossing in the uncapped arms occurs at tick 8. A careful trace review compares the state immediately before and after that event. Tick 7 still has the source on the non-crossed side of the boundary. At tick 8, accepted withdrawal plus local drive places the successor below 11. The crossing counter increments once; remaining below does not create a new crossing every tick.

In P1C, the same growing request meets a shrinking safe budget. The scaling factor falls below one, accepted export is reduced, and the after-state reaches 11 without passing it. On later ticks the source’s regeneration and accepted export balance at the boundary. The trace therefore records repeated binding, not repeated crossing.

This comparison helps readers interpret three different metrics:

- *crossing count* records entry through a boundary;

- *time below* records persistence in the harmful region;
- *minimum stock* records the deepest excursion.

One metric cannot replace the others. A single crossing followed by long recovery is different from a crossing followed by 193 ticks below reserve.

55.6 Why “dead source” is not the only collapse indicator

The two collapsing arms finish with negative source states in the unconstrained model, yet their registered “dead source” boolean remains false because that diagnostic used a different terminal definition. The record therefore uses reserve crossing, Allee crossing, time below thresholds, physical overuse, and domain status rather than relying on one boolean.

This is an important general lesson. A biological system can lose regenerative capacity before its stock equals zero. A hospital can remain open while losing critical service capacity. Collapse is often a process with several observable boundaries, not one dramatic final event.

55.7 What D9 proved and observed

The P1C reserve inequality is an algebraic one-step result under stated assumptions. D9 additionally observed that the implementation held the source at the reserve for this 200-tick fixture, with no observed theorem violation. It also observed that the soft penalty did not preserve the source and that the cap created explicit rationing.

D9 did not prove long-run sustainability, correct reserve certification, social fairness of proportional rationing, stochastic robustness, or a general service theorem. It demonstrated a clean distinction between warning and constraint.

55.8 A needs interpretation

Imagine the destination is a hospital rather than an anonymous sink. The 69.175 units of unmet demand are serious; preservation does not make them disappear. A responsible institution would search for another source, substitute treatment, borrow emergency capacity, improve route efficiency, or ex-

pand regenerative supply. What it should not do is erase the source boundary from the account and call destructive delivery affordable.

P1C makes the scarcity visible early enough for those institutional responses to be designed. It is a physical safety layer, not the complete ethical decision.

Chapter recap

D9 makes the reserve boundary reachable. The soft penalty delays one deeper crossing but does not prevent the reserve crossing. P1C holds the source at 11 with zero observed violations and reports 69.175 units of unmet demand. The figure and table must be read together: preservation, delivery, and scarcity are separate outcomes.

Exercises

1. Why does the C-versus-D comparison isolate the hard cap from the soft reserve weight?
2. Is arm A's larger delivery evidence that it is the best long-run policy? Explain using the threshold crossings.
3. Propose two institutional responses to D9's unmet hospital need that do not redefine the local source physics.

D10: a map of service, rationing, and collapse

One carefully chosen trajectory can still be exceptional. D10 varies demand, transport efficiency, and several secondary parameters to map where the four policies preserve, serve, ration, or collapse. The purpose is not to simulate a planet. It is to see whether the D9 mechanism persists across a registered grid and whether the outcome labels behave sensibly near analytic boundaries.

56.1 The two physical ratios

A logistic source has a maximum regeneration rate $g_{\max}$. Demand is expressed as the dimensionless ratio $d/g_{\max}$. A value below 1 means demand is below maximum source regeneration before transport loss. Useful delivered capacity is smaller when $\eta < 1$, so the more relevant service boundary is approximately

$$d \leq \eta g_{\max}. \quad (56.1)$$

This creates a region where the source could regenerate enough in gross terms but the route cannot deliver enough after loss. Transport efficiency is therefore not merely a cost modifier; it can move a need from feasible service into physical scarcity.

56.2 The registered grid

The D10 core crosses demand ratios 0.25, 0.5, 0.9, 1.0, and 1.1 and efficiencies 0.5, 0.7, 0.9, and 1.0. It compares P0, P1, soft, and P1C. Secondary slices vary activation threshold θ , reserve margin, soft weight χ , regeneration rate ρ , and timestep ratio. The complete D9/D10 study contains 144 runs; D10 contributes 140.

56.3 How to read the phase map

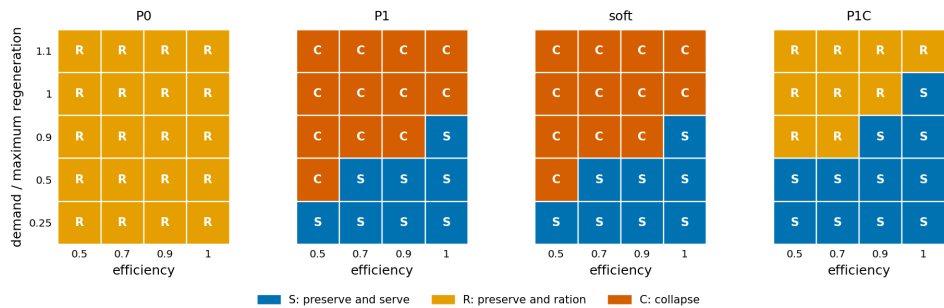


Figure 56.1: *D10 core outcome map generated from the committed summary. Each panel uses the same demand and efficiency axes. S means safe service, R safe rationing, and C collapse.*

Each small square is one registered run. Move right to improve delivery efficiency. Move upward to increase demand relative to maximum regeneration. Compare panels horizontally to ask how policy changes the outcome at the same physical point.

The P0 panel is entirely rationing because no transport serves the destination. Calling this panel “safe” would be misleading: source preservation is achieved by doing nothing for the destination. P1 and the soft arm serve the lower-demand, higher-efficiency region, but collapse through overuse at 25 of 35 registered points. P1C has no collapse points. It safely serves 25 and safely rations 10.

56.4 Outcome counts

Policy	Collapse	Safe service	Safe rationing
P0	0	0	35
P1	25	10	0
soft	25	10	0
P1C	0	25	10

P1C does not manufacture capacity. Where demand and transport make full service infeasible under preservation, the map turns from S to R. This is why

explicit unmet demand is scientifically useful. It shows the boundary between a policy failure and a physical impossibility within the declared model.

56.5 Reading one vertical and one horizontal slice

Choose the P1C panel and hold efficiency fixed at 0.9. Moving upward increases demand while route quality stays constant. Lower rows remain safe service; at higher demand the cells become safe rationing. The transition says that the same route and controller meet small needs fully but reveal scarcity when the required flow approaches or exceeds sustainable useful capacity.

Now hold demand ratio at 0.9 and move horizontally. Increasing efficiency means more of each source withdrawal reaches the destination. The service region expands because less source stock is spent per useful delivered unit. This is the kind of structural incentive an EBU account can expose: improving a route can change feasibility, not merely reduce an arbitrary charge.

Compare the same square across panels. If P1 collapses while P1C rations, the difference is not demand or efficiency; those are held fixed. It is the hard source constraint. If P0 rations while P1C serves, the difference is the ability to transport useful stock while respecting the reserve.

56.6 Why a colour map needs letters and counts

Colours provide a fast visual pattern but can be inaccessible or ambiguous. The map therefore places S, R, or C inside every cell and the text supplies exact counts. A reviewer should be able to recover the conclusion without relying on colour perception.

The count table also prevents visual area from becoming an informal statistic. The registered grid contains 35 points per policy, so “25 safe service and 10 safe rationing” is an exact inventory, not an estimate from coloured pixels.

56.7 Matching service where service is feasible

A meaningful preservation controller should not ration every world merely to make the reserve plot look safe. In the feasible region, P1C matches the serving arms’ useful service while avoiding the collapse region. At higher demand or lower efficiency it reports safe rationing.

This is a finite grid observation, not a universal optimality theorem. Yet it answers an essential diagnostic question: the hard cap is not equivalent to always refusing transport.

56.8 Secondary slices

The secondary parameter slices ask whether one headline map is hiding a fragile choice.

Activation threshold θ . A larger threshold requires a stronger field force before the route activates and can reduce service near balance.

Reserve margin. Raising the effective reserve reduces the stock available for export and can move a point from service to rationing.

Soft weight χ . Changing the warning strength modifies the field force but does not create the hard invariant seen under P1C.

Regeneration rate ρ . Faster regeneration expands the physically feasible region; slower regeneration contracts it.

Timestep ratio $r_{\Delta t}$. A step nearer the certificate probes finite-step sensitivity without pretending the ratio is a real-world safety margin.

The slices are interpretation aids. They do not exhaust the parameter space or replace an uncertainty analysis.

56.9 Domain exits remain results

Eighteen D10 trajectories end with a domain-exit status. They are not removed from the denominator or silently clipped. A domain exit tells the reader that the unconstrained numerical model left the declared physical region. Its later service values are not treated as valid physical delivery.

Dropping failed runs would make the harmful policies look better precisely where their dynamics become impossible. Keeping them is part of honest failure reporting.

56.10 The serialization incident

The first D9/D10 execution attempt failed at the artifact layer. Fifteen stability diagnostics overflowed to non-finite values that strict JSON could not represent. The trajectories themselves had already run. Under a separately documented narrow repair, those diagnostic fields were reconstructed from committed traces and represented as null; no trajectory, service total, crossing, or outcome class was rerun.

The incident distinguishes three operations that are often blurred: executing a model, interpreting a trace, and serializing a report. A failure in the third does not justify secretly repeating the first.

56.11 What the map supports

D10 supports the finite observation that P1C avoided collapse across the registered grid, served the feasible region, and exposed scarcity as rationing. It also shows that P0's preservation is socially empty in these worlds and that the soft penalty did not replace a hard cap.

The map does not prove that every real resource has logistic dynamics, that the declared reserve is correct, or that proportional sharing is fair. It does not close the theorem gap for the bounded service wrapper. A phase map is a readable summary of finite registered runs, not a law of nature.

Chapter recap

D10 maps demand and efficiency against four policies. The map must be read as a joint preservation-and-service instrument: P0 rations everything, P1 and soft serve some points but collapse at many others, and P1C moves between safe service and explicit rationing without observed collapse. Domain exits and serialization history remain visible.

Exercises

1. For $g_{\max} = 3$ and $\eta = 0.7$, what is the maximum useful delivery rate before other constraints?
2. Why is P0's all-rationing panel not a preservation success by itself?

3. Choose one D10 square and describe what additional evidence would be needed before treating it as a real resource-management recommendation.

Gate 1B: does the quote code match the equation?

The V2.9 studies established a preservation controller, but a future action account requires another object: a signed quote that is exact for the represented local transformation. Gate 1B did not ask whether actors following that quote would behave well over time. It asked a narrower question first: does the pure quote implementation conform to the preregistered mathematical definition?

No behavioural trajectory was run. The gate consisted of 21 conformance groups, Q1 through Q21, plus a read-only audit. The adversarial replay Q22 was already named in the locked plan but deliberately reserved for a later authorization.

57.1 Why equation conformance comes before behaviour

If a behavioural study produces a surprising outcome, the investigator needs to know whether the actor followed the intended quote or a coding error. Separating equation conformance from behaviour reduces that ambiguity. Gate 1B therefore tests the quote as a mathematical instrument before placing it inside a selection loop.

This is similar to calibrating a thermometer before using it in a clinical trial. A calibration result does not prove that a treatment works. It establishes that the measurement device reports the temperatures it claims to report at the tested points.

57.2 Q1–Q5: the five basic stories

The first groups use human-readable fixtures.

Q1: doing nothing

At $q = 0$, no represented state changes and $C_a(0) = 0$, so the exact quote must be exactly zero. This checks the explicit zero branch rather than a small positive limit.

Q2: a beneficial action

Worked example E1 has exact quote +7.94. The linear diagnostic reports +15.18. Both are positive, but the linear tangent substantially overstates the finite benefit. The exact value is the settlement result.

Q3: a damaging action

Worked example E2 has exact quote -5.52 and linear diagnostic -0.08 . The linear approximation misses 5.44 burden units. The test contains a negative control showing that a settlement path using the diagnostic would be detected.

Q4: a closed damage-repair cycle

In E3, a sequence returns the state exactly to $(10, 10)$. The net state-field difference therefore telescopes to zero. The remaining settlement is the negative sum of declared process costs, -0.10 . A closed physical state cycle cannot manufacture positive EBU merely by splitting the path.

Q5: natural regeneration is not actor credit

The regeneration fixture has drive $g = 2$ and no-action successor $z = 12$. If the actor does nothing, the quote is exactly zero. A deliberately naive baseline using the pre-drive state credits or debits the actor for natural regeneration; the test detects the error exactly.

Together these cases teach the quote's grammar: rest is zero, signed action values can be positive or negative, finite curvature matters, cycles pay process burden, and background drive belongs in the counterfactual baseline.

57.3 Q6–Q8: curvature and the complete schedule

Q6 compares exact and linear values across penalty branches and 200 registered random cases. For a convex state penalty and the declared cost family, the first-order tangent provides an upper diagnostic bound on the exact benefit. The group checks the numerical implementation of that relation at the sampled points.

Q7 studies schedule concavity and the discontinuity created by a fixed setup cost at $q = 0$. “Concave schedule” here means that, within the registered cost assumptions, increasing quantity has diminishing exact benefit. It does not mean that every larger quantity remains beneficial.

Q8 commits complete schedules for 100 cases and evaluates five partial-execution points on each. The test checks that settling a smaller measured quantity uses the already committed function and remains inside $[0, q^{\text{acc}}]$.

57.4 Q9 and Q12: protocol violations are not valuations

Q9 requests settlement beyond the accepted maximum. The minimum fail-closed envelope issues no positive EBU and creates a mandatory violation record. Q12 attempts duplicate settlement on ten fixtures and requires rejection.

The important design choice is that the violation layer does not invent a field value. If a shipment exceeds its authorization, the system has at least two facts: the physical action occurred, and the protocol was violated. A security or legal response may be necessary, but its penalty should not masquerade as the exact field difference.

57.5 Q10 and Q11: telescoping with and without drive

For a sequence of exact action differences evaluated on consistent successive states, intermediate state terms telescope. In an undriven closed cycle, the net state part is zero and process burden remains. Q10 checks 50 sequences and ten cycles.

With natural drive, a separate residual appears because the no-action successor changes between epochs. Q11 checks the driven identity and includes a negative control that wrongly applies the undriven theorem; the omitted residual exceeds 10^{-3} and is detected. This prevents a concise conservation slogan from being used outside its assumptions.

57.6 Q13 and Q14: the split-action trap

The registered split counterexample evaluates two withdrawals from the same frozen nonlinear source. Quoting them independently as if each were first gives a sum of 32, while the joint state change is 20. The phantom credit is 12.

The cause is curvature. Both independent lines claim the high marginal benefit available before either withdrawal. In the joint action, the first withdrawal changes the source state seen by the second part of the exact finite difference.

Q14 therefore enforces a first production discipline: one quoted action per source per micro-step. This avoids issuing independent overlapping claims while a general aggregate allocation rule is unproved. The restriction is conservative for accounting but can reduce action capability; later gates measure that cost.

57.7 Q15–Q17: permission and observation

Q15 constructs an action with a positive raw quote beyond P1C permission. The quote module refuses to extend the schedule past q^{acc} . Q16 constructs a P1C-permitted action with negative EBU, demonstrating the converse: permission does not promise a positive reward.

Q17 compares a non-binding P1C physical step with and without observational quoting. The physical tuples are exactly identical. This fixture is small, but it establishes the intended non-interference path before the same idea is tested over full trajectories.

57.8 Q18–Q21: the boundary around the equation

Q18 scans imports and forbidden identifiers, injects runtime poison, and rejects a world object where only two local views are permitted. Q19 rejects

process-cost categories that would double count state burden or mix money, labour, and security penalties into physical EBU. Q20 tests stale epochs, mismatched commitments, and the rule against reallocating a quote to another action. Q21 rebuilds 52 objects twice, checks deterministic identifiers, and requires strict finite JSON.

These groups test not only a numerical formula but its operational boundary. A quote is scientifically meaningful only if the system knows which data it may read, which state it belongs to, and whether it can be replayed.

57.9 Random fixtures without behavioural randomness

Some conformance groups use pseudorandomly generated input points under frozen ranges and seed 30001. This is not a stochastic behavioural study. The seed makes the input set exactly reproducible, while varied points reduce the chance that a branch error hides behind one hand-picked example.

The registered ranges include stocks from 0 to 24, several penalty weights, reserve values, efficiencies from 0.5 to 1, two timesteps, positive and zero setup costs, and drives between -2 and 2. Non-vacuity checks require the resulting sample to exercise more than the zero branch.

57.10 Audit independence

The separate audit recomputes plan hashes, checks the implementation import allowlist, verifies that the test does not import the adversarial runner, confirms equation-version consistency, reconstructs epoch identifiers, checks strict JSON, and verifies that released V2.9 files did not change.

This audit does not repeat every unit test. It attacks the conditions under which the test result should be believed. An implementation can return correct fixture numbers yet still be unacceptable if it imports global actor state or silently modifies the released physics.

57.11 What passed and what remains open

The conformance suite reported 184 passing checks in 21 groups, and the audit reported 19 passing checks. The maximum residual in the named worked

examples was approximately 8.9×10^{-16} , well inside the declared scale-aware tolerance.

The result supports a precise statement: at the declared fixture points, the pure quote implementation behaved consistently with the locked exact equation and its protocol boundary. It does not establish incentive compatibility, adversarial security, long-horizon alignment, robust observation under measurement error, or a complete actor economy. Those are different gates.

57.12 How the groups cover the action lifecycle

The 21 groups are easier to remember as a lifecycle than as a numbered list.

1. **Define value:** Q1–Q7 check zero, signs, cycles, drive, curvature, and schedule shape.
2. **Commit an opportunity:** Q8 binds the full schedule and permits partial execution inside its domain.
3. **Handle invalid execution:** Q9 and Q12 record overexecution and duplicates without issuing invented value.
4. **Compose time:** Q10 and Q11 distinguish undriven telescoping from driven residuals.
5. **Protect shared sources:** Q13 and Q14 expose split over-credit and enforce the conservative micro-step rule.
6. **Respect physical permission:** Q15–Q17 test P1C domain, negative permitted actions, and observational identity.
7. **Protect meaning:** Q18–Q21 enforce information, cost, epoch, and serialization boundaries.

The next gate begins only after this lifecycle is coherent. Otherwise a later behavioural failure could be caused by a malformed schedule, replayed epoch, or double-counted cost rather than the actor's selection objective.

57.13 A reader's reconstruction of E1

To reconstruct a worked quote without reading Python:

1. write the two endpoint states and local drives;
2. calculate z_i and z_j ;
3. evaluate both local penalties at z ;
4. apply $-\Delta tq$ at the source and $+\Delta tq$ at the destination;
5. evaluate both penalties again;
6. subtract the after penalties and $C_a(q)$ from the before penalties;
7. separately calculate the linear diagnostic and compare, but never replace the exact value.

This sequence is the conceptual core of the implementation. Hashes, epochs, and registries make the same calculation auditable in a multi-action program.

Chapter recap

Gate 1B calibrated the quote layer rather than testing an economy. Its fixtures cover zero, beneficial and damaging actions, cycles, drive, curvature, partial execution, violations, split-action over-credit, P1C boundaries, information boundaries, epoch integrity, and serialization. Passing results support implementation conformance at declared points; they do not answer behavioural alignment.

Exercises

1. Explain the phantom 12 in the split counterexample without using the word “bug.” Which mathematical property creates it?
2. Why does Q15 need an action that is both physically forbidden and economically attractive under the raw quote?
3. Select three Q-groups and classify each as equation, protocol, or information-boundary evidence.

From adversarial search to service alignment

After calibrating the quote, the project asked a more dangerous question: can an actor select locally profitable exact quotes while producing persistent physical harm? Gate 1C searched for such behaviour. It did not find the registered exploit, but it produced a different surprise – the quote-maximising arm served substantially less than the ordinary P1C baseline while still earning positive EBU. Gate 1D was designed to understand that surprise.

58.1 Gate 1C as a falsification search

Gate 1C translated twelve released 5-by-5 layouts into the V2.9/P1C substrate. For each layout it ran three production arms:

Arm A. P1C physical baseline, no EBU-guided selection.

Arm B. The same P1C actions with observational exact quoting.

Arm C. A production-local adversary that evaluates its permitted local menu and chooses the action with the greatest exact quote.

A separate red-team oracle used global harm as an external search objective. It was labelled as a testing instrument, never as information available to a production actor. The frozen search used depth 10, width 40, a 20-tick tail, two quantity fractions, a two-actor coalition, and layout seeds 0 through 11.

58.2 The registered exploit predicate

The standing falsifier required positive cumulative exact EBU together with persistent harm relative to a no-action baseline. This is a conjunction: profit

alone is not the exploit, and harm alone is not the exploit. The tail condition requires persistence rather than one noisy tick.

No registered production run or oracle search satisfied the predicate. Historical positive controls did. The released V2.6 naive ledger reproduced an exploit with net EBU +15.35 and positive persistent harm. The V2.6 seed-0 guarded standing falsifier reproduced +260.13 with five of five sources dead. These controls show that the search machinery could find known failures.

Under P1C and the exact quote, all 36 production runs had zero reserve crossings, zero Allee crossings, zero dead sources, and zero physical overuse. The old harm channel was physically blocked by the permission layer.

58.3 The encouraging result and its boundary

It is tempting to summarise Gate 1C as “no exploit, therefore safe.” The artifact explicitly forbids that claim. The search covered particular layouts, seeds, menu, depth, width, and predicate. It was not a proof of security.

Three limitations were especially serious:

1. all 36 production runs eventually left the physical domain because standing demand drained unconstrained consumer cells below zero;
2. the inherited unit timestep gave $r_{\Delta t} = 2.283 > 1$, outside the discrete certificate;
3. the no-action-relative harm predicate could not detect degradation relative to an actively serving baseline.

The third limitation revealed the central surprise.

58.4 The metrics answer different questions

Gate 1C reports several columns because no single one defines harm.

Cumulative EBU. Sum of settled exact action lines in the observational research account. It is not a wallet balance in this gate.

Final burden. The global diagnostic V at the last recorded state. It can compare represented field condition but does not directly measure service.

Viability percentage. Fraction of cells inside their declared viable bands. Two worlds with the same percentage can contain different cells outside the band.

Served quantity. Useful demand met before physical invalidity. It does not by itself reveal source depletion.

Reserve and Allee crossings. Entry into named source-danger regions.

Physical overuse. Accepted withdrawal beyond the P1C budget, which remained zero.

Domain exit. A state leaving its declared physical interval. This limits interpretation of all later ticks.

Persistent harm. A registered tail comparison with a baseline and margin, not a synonym for any one bad final number.

Arm C can therefore have positive cumulative EBU, zero source crossings, worse service, higher final burden, and a false persistent-harm predicate at the same time. The statements are not logically inconsistent; they refer to different comparisons. The scientific task is to decide whether the predicate captures the failure mode the gate claims to search for.

58.5 Why the no-action baseline was attractive

A no-action baseline provides a clear counterfactual: did acting make the world worse than doing nothing? It is useful for detecting destructive action in a static world. In a driven needs system, however, doing nothing can be extremely harmful because demand continues. A serving policy can degrade substantially and still remain better than total inaction.

The lesson is not that no-action baselines are always wrong. It is that baseline choice is part of the scientific claim and should be preregistered alongside a positive control showing what the predicate can detect.

58.6 Positive EBU with worse service

Averaged across twelve layouts, arm A served 247.3 units, ended with burden 1466.1, and final viability 50 percent. Quote-maximising arm C earned mean

EBU +243.60, but served only 166.0, ended with higher burden 1561.7, and final viability 42.3 percent.

The registered predicate still reported no exploit because both serving arms were far better than doing nothing. In symbolic terms, if no-action harm is very high, both A and C can satisfy

$$\text{harm}(A) < \text{harm}(C) < \text{harm}(\text{no action}), \quad (58.1)$$

so a test comparing only C with no action misses the fact that C is worse than A.

The predicate was not rewritten after seeing the result. Instead, the surprise created two open questions: a serving-baseline predicate, and whether the apparent service loss came from the quote objective or from unequal action capability.

58.7 The capability confounder

Arm A could accept many outgoing edge flows from one source during a tick. To avoid the split-action phantom-credit problem, arm C proposed only one quoted action per source per micro-step. Observational arm B could quote only a small minority of all accepted flows for the same reason. Thus the comparison changed two things at once: selection objective and available action capability.

If A serves four destinations simultaneously while C chooses one, lower service may be caused by the restriction even when C chooses the best possible single edge. Gate 1C could not separate those explanations.

58.8 Seed 0 as a concrete comparison

The standing seed-0 layout makes the ambiguity tangible. Plain P1C arm A served 120.57 units and ended with burden 3268.51. Observational arm B produced the same physics exactly and settled only one quotable action for +0.309 EBU. The quote-maximising arm C settled 140 actions for +196.564 EBU, served 108.28, and ended with the higher burden 3354.61. None crossed a reserve or killed a source.

The result contains both progress and concern. P1C prevented the catastrophic source-death exploit seen in the historical V2.6 control. Yet maximising

available quotes did not reproduce the serving baseline. Because C also had fewer action opportunities, the trace cannot assign that gap uniquely to the quote objective.

This is why a good experiment does not stop at “falsifier false.” Secondary metrics can reveal that the predicate asks too weak a question.

58.9 Coverage is part of capability

Observational arm B could quote only 0 to 43 actions per layout while hundreds of P1C edge flows occurred. The remaining flows were labelled `unquotable_multi_edge`; they were not silently assigned zero EBU. The small coverage fraction follows from the shared-source accounting restriction, not from missing trace data.

Gate 1D later constructs full quote coverage by restricting every compared arm to the same one-action menu. That removes one confounder but also changes the physical capability being compared with full P1C. A-versus-B and B-versus-D are therefore both required.

58.10 Gate 1D redesigns the service instrument

Gate 1D introduced a bounded service wrapper. Demand was removed from the unconstrained drive and applied after transport:

$$\text{available}_j = x_j + \Delta t \left(u_j + \sum_{e \rightarrow j} \eta_e q_e^{\text{acc}} - \sum_{e \leftarrow j} q_e^{\text{acc}} \right), \quad (58.2)$$

$$\text{service}_j = \min(\Delta t d_j, \text{available}_j), \quad (58.3)$$

$$\text{unmet}_j = \Delta t d_j - \text{service}_j, \quad (58.4)$$

$$x_j^{n+1} = \text{available}_j - \text{service}_j \geq 0. \quad (58.5)$$

This wrapper prevents phantom consumption. If only 0.6 stock units are available and demand is 1, service is 0.6 and unmet demand is 0.4. The state ends at zero, not negative 0.4.

The wrapper is a registered semantics extension and lies outside the original D0 theorem. Its non-negativity and ledger properties are tested; a general constrained theorem remains open.

58.11 Four capability-matched arms

Gate 1D used:

- A: full P1C capability;
- B: restricted P1C using a non-EBU force selection;
- C: the same restricted physical action plus observational quote;
- D: the same restricted menu with exact-quote-greedy selection.

B is the primary baseline for D because both may select the same number of actions from the same menu. A-versus-B measures the price of the restriction. B-versus-C tests observational identity. B-versus-D asks whether replacing force ranking with exact-total-quote ranking harms service under matched capability.

The study used seven small worlds, two certified timesteps per world, four arms, 200 ticks, and a 50-tick burn-in: 56 runs in total. All paired timesteps satisfied $r_{\Delta t} \leq 1$.

58.12 What Gate 1D observed

Bounded service removed every domain exit: zero of 56 runs had a negative state. Service never exceeded available stock or demand, unmet demand closed exactly, physical overuse was zero, and the maximum ledger residual was 8.77×10^{-15} .

B and C were byte-identical in all fourteen world-timestep pairs. B and D also selected the same physical action at every tick. Their post-burn-in service was identical, with zero service-alignment failures. D accumulated positive EBU, but the quote and force rankings agreed on the only active choice.

This was a cleaner result than Gate 1C, but it contained another instrument problem: every source had at most one outgoing edge. The one-action restriction could not bind. A and B therefore had identical capability, and the study could not measure the very confounder it was designed to quantify.

58.13 The one-ULP diagnostic defect

The first Gate 1D rendering labelled four W2 conservative runs as reserve crossings because the minimum stock was 7.999999999999999 against reserve 8. The difference was one unit in the last place, approximately 8.88×10^{-16} . The crossing counter used a strict comparison without the registered scale-aware tolerance.

The original attempt was preserved. Under a narrow new authorization, Gate 1D-A changed only the diagnostic predicate and generated a second rendering. The uncompressed traces of all 56 runs were byte-identical. Only four crossing labels, their derived outcome classes, and associated diagnostic flags changed. No material reserve crossing remained, and the B-versus-D result stayed exactly the same.

This correction is a model of how to repair labels without pretending the first attempt never happened.

58.14 What was learned before O14

Gate 1C showed that P1C blocked the historical source-destruction exploit but also revealed a service comparison confounded by capability, domain exits, uncertified time, and a blind predicate. Gate 1D removed the domain and timestep confounders, implemented a serving baseline, and found no misalignment under matched capability. Yet its worlds did not contain genuine destination choice, so it could not test whether force and exact quote rank several active outgoing edges differently.

The next study therefore needed sources with two to four simultaneously active outgoing edges, a restriction that demonstrably binds, and worlds where B and D are analytically predicted to choose different destinations. That study was O14.

Chapter recap

Gate 1C found no registered persistent-harm exploit, but its quote-maximising arm served less than the P1C baseline and exposed weaknesses in domain semantics, timestep certification, and the outcome predicate. Gate 1D bounded service, matched capability, and found no service difference, but

its one-out-edge worlds made genuine choice impossible. Gate 1D-A corrected only four floating-point labels. The sequence shows experiments refining instruments rather than moving directly from one headline to another.

Exercises

1. Draw the inequality that makes the no-action predicate blind to C's degradation relative to A.
2. Why is B, not A, the primary baseline for D in a capability-matched test?
3. Explain how the byte-identical Gate 1D-A traces distinguish a diagnostic repair from a changed scientific trajectory.

O14: genuine destination choice and a flat outcome

O14, also called Gate 1D-B, was designed around the limitation left by Gate 1D. Every registered source now had between two and four active outgoing edges. The study could observe real destination choice, a genuinely binding multi-action restriction, per-unit-versus-total quote disagreement, and aggregate shared-source over-credit.

Its headline result was not the expected decisive separation. All six worlds were non-discriminating under the registered primary outcome. Falsifier F13 fired. Understanding why requires reading behaviour, service semantics, and the outcome instrument together.

59.1 The six worlds as six questions

Each deterministic world contains one configured source and a static graph.

- W1: efficiency split.** Can route efficiency change choice, and can a stock-blind delivered-volume heuristic neglect a vulnerable destination buffer?
- W2: severity split.** Can force and exact total quote provably prefer different destinations from the first tick?
- W3: volume split.** Can a smaller material action produce more EBU than a larger delivered-volume action?
- W4: budget binding.** What happens when four simultaneous outgoing requests share a source only 0.6 stock units above reserve?
- W5: reversal.** Will the preferred edge change over time as local deficits move?
- W6: infeasibility.** Will the system expose demand greater than useful reserve-limited capacity as explicit unmet need?

Static algebra established non-vacuity before execution: at least two edges had positive raw flux at the initial state of every world, W4’s aggregate budget had to bind, W6 demand exceeded the useful maximum, and W2 contained a predicted B-versus-D destination divergence.

59.2 What would count as informative in each world

The world names describe mechanisms, but each also has a specific observable.

World	Designed tension	Key evidence to inspect
W1	high- versus low-efficiency destinations	service by destination and S-arm buffer exhaustion
W2	force prefers severe-small while quote prefers mild-large	first selected edge and accumulated B/C/D quote totals
W3	delivered volume and exact burden improvement disagree	which heuristic under-serves and whether less material earns more EBU
W4	four requests share a near-reserve source	σ_A , budget utilisation, action count, and A/B post-burn-in service
W5	local ranking reverses as destination states evolve	selected-edge switch counts and timing
W6	demand exceeds useful maximum	unmet demand, zero reserve crossings, and physical-impossibility class

This table prevents a common mistake: judging every world only by total EBU. W4 is primarily a capability test; W5 is an adaptation test; W6 is an infeasibility test. The same trace fields have different evidential roles in each fixture.

59.3 Five arms and the primary comparisons

The arms were:

- A. Full simultaneous multi-edge P1C. It measured physical capability and recorded an aggregate quote diagnostic, but settled no EBU.
- B. Restricted matched non-EBU force selector. This is the primary baseline.
- C. The same physical selector as B with observational quotes.
- D. Restricted exact-total-quote greedy selector. It rests unless the best exact quote is strictly positive.
- S. A restricted local service-volume comparator, intentionally stock-blind and used only for secondary attribution.

Every restricted arm consumes the same candidate menu. Selectors do not build their own menus. Request shaping makes the selected accepted quantity equal to the quantity executed by the unchanged P1C step. These details prevent a hidden capability difference from returning through implementation.

59.4 Why total quote, not EBU per unit

The registered heuristic ranks complete candidate actions by $\Delta_e(q^{acc})$. It does not divide by quantity. Per-unit ranking answers “which unit is most efficient?”; total ranking answers “which available action produces the largest positive field improvement?” Those are different objectives.

O14 observed 4,798 candidate-ranking ticks where per-unit and total rankings diverged. The selector nevertheless followed the exact total rule. This makes the distinction operational rather than a footnote.

Under the registered costs and certified timesteps, the exact schedule was strictly increasing over the feasible four-level quantity menu in every world. Quantity misalignment was therefore structurally excluded in this design; the scientific choice lay mainly between destinations. The continuous schedule vertex was recorded only as a diagnostic and never used by the selector.

59.5 One restricted-arm tick from menu to service

At the beginning of a restricted-arm tick, the implementation performs the following sequence:

1. freeze the world state and calculate local drives;
2. calculate the authoritative P1C budget of the configured source;
3. calculate raw D0 flux on every outgoing edge;
4. form four registered quantity fractions for each edge, capped by the same budget logic;
5. present the identical candidate dictionary to B, C, D, and S;
6. let the arm selector choose one candidate or rest;
7. shape the active world so unchanged `p1c_step` executes the selected accepted quantity exactly;
8. apply bounded destination service and record unmet demand;
9. calculate quotes and global diagnostics only in their registered layers;
10. append one complete tick record.

Arm B orders candidates by loss-aware force, then accepted quantity, with identifier tie-breaking. Arm D evaluates each complete exact quote and chooses the largest strictly positive value, also using identifiers only for exact ties. Arm S maximises useful delivered volume toward a demanding destination without reading destination stock. Arm C selects exactly as B and merely observes the quote.

A test drops one candidate from one arm's menu and requires detection. Another test constructs a tie in both identifier directions. These controls ensure that the observed W2 choice divergence comes from the registered score, not from menu construction or iteration order.

59.6 Why selected quantity had to equal executed quantity

If the selector quotes one quantity but P1C later rescales it differently, the study would compare a planned score with another physical action. O14 avoids that confounder by using the same arithmetic as P1C when constructing candidates and by shaping the request so the selected q^{acc} is the executed value.

Tests prove bit-identical equality on all W1 candidates at both timesteps. An unscaled negative control executes a different quantity and is detected. The identity is also asserted on every official tick before the run can complete.

59.7 The registered study and execution integrity

The Cartesian product was six worlds, five arms, and two timesteps: 60 runs. Every run lasted 200 ticks with a 50-tick burn-in and 20-tick persistence window. The trace contains exactly 12,000 ordered tick rows. Every row carries the canonical plan hash. No seed, retry, filter, subset, or rerun was used.

All scientific numbers were finite. There were zero domain failures, no negative state, no material reserve or Allee crossing, and no dead source. Aggregate service, EBU, unmet demand, and crossings reconstruct from trace rows within 10^{-9} ; the maximum ledger residual was approximately 1.03×10^{-14} .

The official scientific execution occurred only after the earlier runner self-match defect had failed closed, been recorded, corrected, re-tested, and separately authorized.

59.8 What behaviour actually differed

B and C were physically identical in all twelve world-timestep pairs, with maximum state difference exactly zero. This is strong evidence that observational quoting remained inert.

B and D were not behaviourally identical. In W2 they selected different destinations from tick 1 at both timesteps. In W5 D switched selected edges 37 and 33 times, while B switched 39 and 35 times. D's accumulated EBU differed from C's in W2, confirming that D did not merely quote B's actions.

Arm A acted simultaneously on multiple edges during 2,399 ticks. W4 and W6 had binding aggregate budgets, with very small scaling factors at their most constrained ticks. The one-action restriction was therefore real.

These observations are crucial because F13 did not fire from absence of choice. Choice and capability differences were present; the primary service outcome did not respond to them.

59.9 The primary null and F13

The primary discriminator marked a world as informative if either A-versus-B capability cost was materially non-zero or D-versus-B post-burn-in service ratio differed materially from 1 at either timestep. None did. The list of discriminating worlds was empty, so F13 fired exactly as registered.

D and B had equal post-burn-in service within the preregistered tolerance in all twelve comparisons. One W6 conservative-timestep ratio was 0.9999999999999999, one ULP below one and inside that tolerance. A and B also had zero post-burn-in capability cost everywhere, including W4 and W6 where the restriction bound. H3, the prediction of a non-zero capability cost somewhere, was not supported.

This is a negative result about the sensitivity of the chosen world-and-outcome instrument. It is not proof that quote-greedy selection and the restriction are harmless.

59.10 How destination buffering creates a plateau

The bounded service wrapper lets a destination store delivered stock. For a non-exporting destination j ,

$$\text{available}_j(t) = x_j(t) + \delta_j(t), \quad (59.1)$$

$$\text{service}_j(t) = \min(\Delta t d_j, \text{available}_j(t)), \quad (59.2)$$

$$x_j(t+1) = \text{available}_j(t) - \text{service}_j(t), \quad (59.3)$$

where $\delta_j(t)$ is efficient delivery during the tick.

If available stock never falls below tick demand during the measurement window, service is exactly the demand cap:

$$S_j(W) = |W| \Delta t d_j. \quad (59.4)$$

Inside this buffered region, changing the delivery schedule does not change cumulative service. Service responds only when a running buffer reaches zero.

The equivalent running condition is

$$x_j(t_0) + \sum_{s=t_0}^t (\delta_j(s) - \Delta t d_j) \geq 0 \quad \text{for every } t \text{ in the window.} \quad (59.5)$$

B and D produced different delivery patterns but stayed inside this inequality. The service functional was flat over the region they occupied.

59.11 Temporal multiplexing

The restriction limits simultaneous actions, not time-averaged throughput. B can serve destination 1 on one tick, destination 2 on the next, and return before either buffer empties. In W4, B selected the four destinations in an exact 50/50/50/50 round-robin. The per-selected-tick useful delivery rate was 2.4075, slightly greater than the total demand rate 2.4. One action at a time was enough to maintain all four stored buffers.

Both B's force and D's exact quote are deficit-seeking. When a destination is not served, its stock falls, its deficit becomes more urgent, and it tends to win a later selection. This feedback behaves like a longest-deficit-first polling loop. The worst B buffer margin in W1 was only 0.368 stock units, so the plateau was thin rather than vacuous.

59.12 Walking through a W4 multiplexing cycle

W4 has four outgoing destinations, each demanding 0.6 stock units per unit time. Arm A can act on all four during a tick. Arm B chooses one. At the initial selected edge, useful delivery rate is approximately

$$\eta J = 0.9(2.675) = 2.4075, \quad (59.6)$$

while total demand rate across all four destinations is

$$4(0.6) = 2.4. \quad (59.7)$$

Suppose B visits destinations 1, 2, 3, and 4 in four successive ticks. A selected destination receives slightly more than four ticks of its individual demand rate, so it can bank enough stock to wait through the other three selections. The trace shows an exact 50/50/50/50 selection count across 200 ticks and 199 switches. The restriction binds instantaneously but not cumulatively at the service read-out.

This explains a result that initially looks contradictory: W4 can have $\sigma_A < 1$, one action per tick in B, and still zero A-versus-B service cost. Instantaneous capability and long-window demand satisfaction are different quantities.

59.13 Hypotheses and falsifiers as two ledgers

The O14 summary reports hypotheses H1–H10 and falsifiers F1–F15 separately. A supported hypothesis records a predicted pattern such as B/C identity, restriction binding, W5 switching, or explicit W6 scarcity. A fired falsifier records a condition declared capable of invalidating or limiting the study.

Nine hypotheses were supported; H3, non-zero capability cost somewhere, was not. F13 fired because no primary world discriminated. The remaining falsifiers did not fire: menus stayed matched, all worlds had several active outgoing edges, information boundaries held, timesteps remained certified, no material reserve harm occurred, and no forbidden post-hoc world was added.

“Not fired” does not mean universally impossible. It means the registered condition was absent from this completed study.

59.14 The S arm reaches the boundary

The stock-blind S comparator served W1 differently. It under-served by 4.315 and 5.000 post-burn-in units at the two timesteps and was classified as distributive-or-policy under-service. S is not part of the F13 primary predicate, so this result does not contradict F13.

It strengthens the buffering diagnosis. The service instrument could respond in the registered worlds when a policy drove a buffer to zero. B and D stayed on the plateau because they tracked deficit; S left it because its score ignored destination stock.

Figure 59.1 should not be read as “all policies are equal.” The left panel shows equality of one saturated outcome for B and D. The right panel shows that another registered heuristic left the saturated region. Behavioural traces are needed to understand why.

59.15 The aggregate quote diagnostic

Arm A permitted simultaneous withdrawals from one nonlinear source. Independent per-edge quotes are not conservative because each line treats the source as if no other withdrawal occurs. Across 2,399 multi-action ticks, the naive independent sum was +3910.750 while the exact group quote was

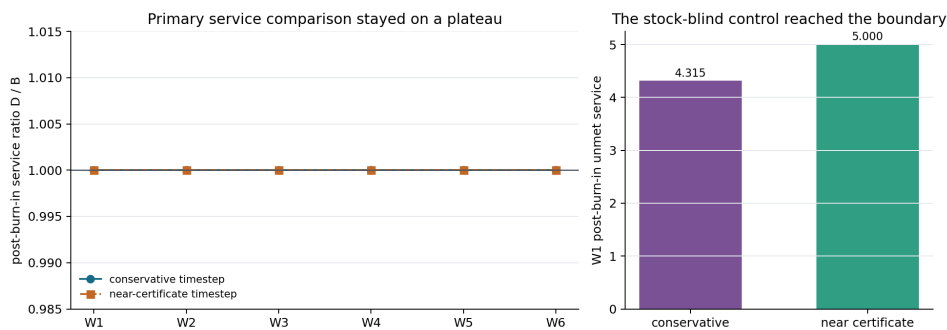


Figure 59.1: *O14 outcome instrument. Left: D/B post-burn-in service remained equal within the registered tolerance in every world; one plotted ratio was one ULP below one. Right: the stock-blind S control produced unmet service in W1, showing that the service boundary was reachable.*

+3894.460. The measured over-credit was 16.289.

Nothing was settled or allocated in arm A. The diagnostic quantifies O3 rather than solving it. Proportional, marginal, source-group, and Aumann–Shapley style allocations each introduce incentive and fairness questions. An arithmetic split that closes is not automatically strategy-proof or scientifically justified.

59.16 Honest infeasibility in W6

W6 demand exceeded useful reserve-limited capacity. All ten W6 runs recorded post-burn-in unmet demand, zero reserve crossings, and the outcome class physical impossibility. Near the reserve the source scaling factor reached zero at the near-certificate timestep.

This is the correct meaning of a controller saying no. The system does not claim that the need is unimportant. It records that the present local field and infrastructure cannot meet it without violating the declared source boundary. That information can motivate another source, storage, route improvement, or capacity construction.

59.17 What O14 changes and what it cannot close

O14 directly measured multi-out-edge capability and genuine destination choice. It found no reserve harm, no B-versus-D service-alignment failure,

and a clear aggregate over-credit diagnostic. It also revealed that the primary outcome was analytically flat over the relevant buffered regime.

O11 therefore remains open. Zero difference on a flat functional is not evidence of alignment outside that plateau. A future outcome-discrimination study needs a positive control and worlds in which delivery timing can exhaust one policy's buffer but not another's. Candidate mechanisms include smaller initial buffers, perishable destination stock, pulsed need, useful flow below multiplexed demand, or service drawn directly from current delivery.

Any such world is a new preregistered study, not a correction to O14. If static algebra cannot construct a discriminating region without post-hoc search, the honest outcome is to stop that line and proceed only with the independent accounting-theory programme.

Chapter recap

O14 created genuine multi-edge choice and binding restrictions. B and D behaved differently but delivered post-burn-in service equal within the registered tolerance because banked destination stock placed the outcome on a plateau. F13 fired as registered. The S control showed that the boundary was reachable, and the arm-A diagnostic measured 16.289 of naive shared-source over-credit without settling it. The result narrows the next question; it does not prove harmlessness.

Exercises

1. Use the running-buffer inequality to explain how alternating two destinations can match simultaneous service.
2. Why does S's W1 under-service strengthen, rather than contradict, the F13 interpretation?
3. Propose a positive-control world in which one selection policy exhausts a buffer and another does not. State which parameter is changed and why the choice is made before running.